short introduction in bio-informatics : data formats and main pipelines

Prof. Laurent Gentzbittel ; Dr. Cécile Ben

École Nationale Supérieure Agronomique de Toulouse – EcoLab

International Course
Plant Quantitative Genetics, GWAS & Genomic Prediction

VIR, July 2018



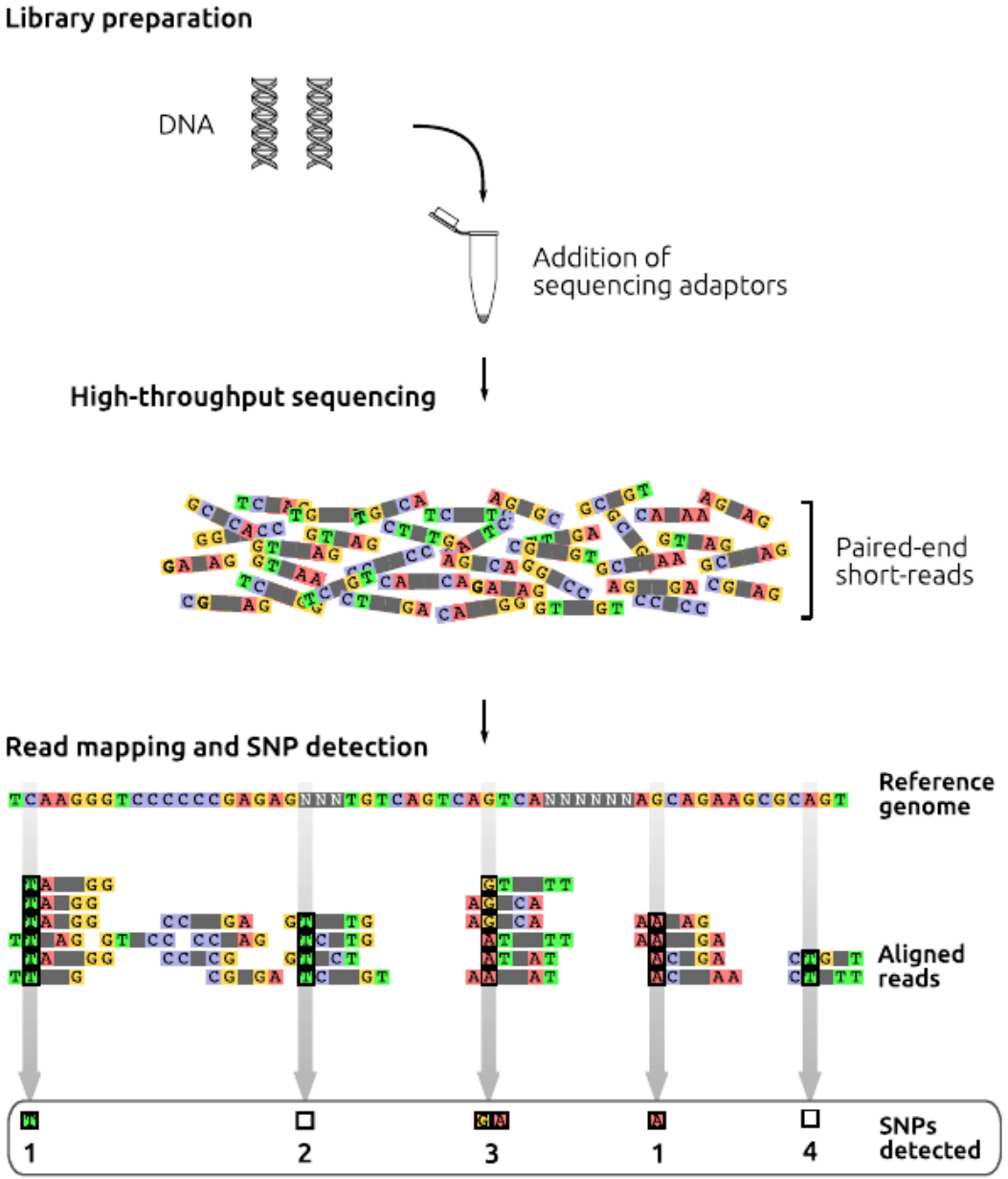
short introduction in bio-informatics

- Detecting SNPs using 'whole genome resequencing'
  - Current tools
  - File/data formats
- Using 'reduced-representation' sequencing libraries (GBS, RAD-Seq , ...) of multiplexed samples
  - GBS with a reference genome
  - GBS w/o a reference genome : UNEAK and STACKS pipelines
- File/data formats for handling SNPs data for subsequent analyses
- quick view : principle of *de novo* sequencing
- What can I do with this genetic data ?

short introduction in bio-informatics

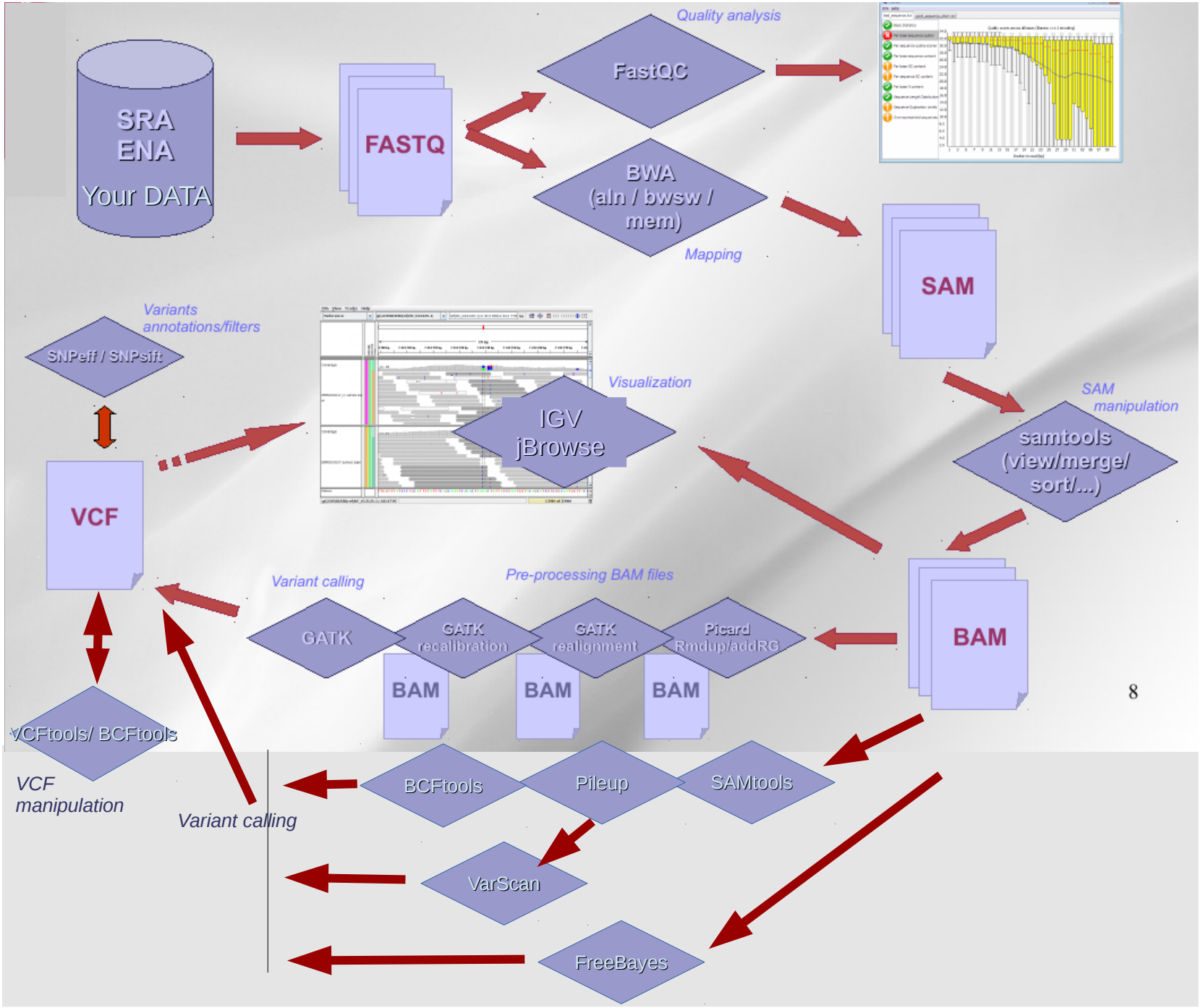
- Detecting SNPs using 'whole genome resequencing'
  - Current tools
  - File/data formats
- Using 'reduced-representation' sequencing libraries (GBS, RAD-Seq , ...) of multiplexed samples
  - GBS with a reference genome
  - GBS w/o a reference genome : UNEAK and STACKS pipelines
- File/data formats for handling SNPs data for subsequent analyses
- quick view : principle of *de novo* sequencing
- What can I do with this genetic data ?

Detecting SNPs with Whole Genome Resequencing



Fuentes-Pardo et al. (2017)
Mol Ecol 26: 5369–5406.

Detecting SNPs with Whole Genome Resequencing

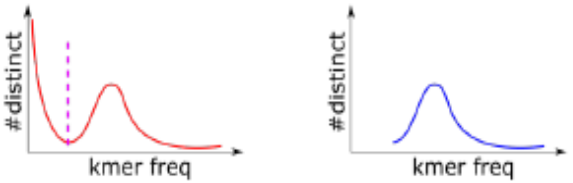


A real workflow for SNP calling

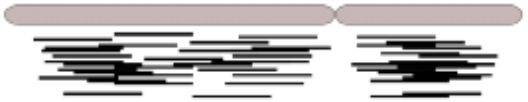
Read clipping



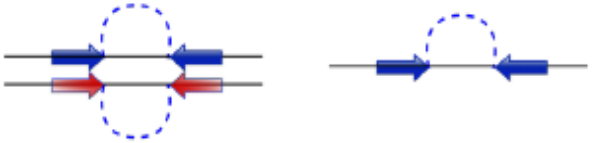
Read error correction



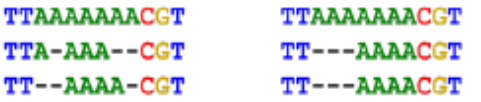
Read mapping



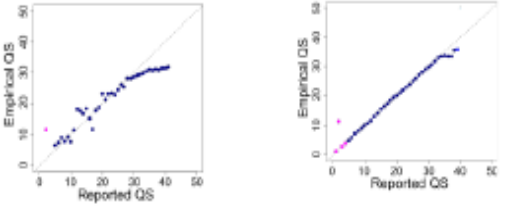
Duplicate removal



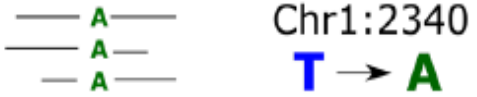
Re-alignment



Base Quality Adjustment

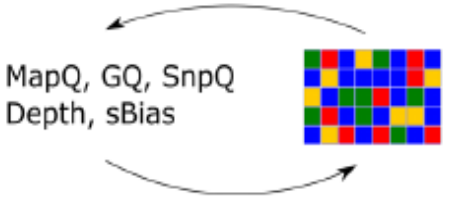


SNP calling



SNP filtering

Evaluation



There are many short-read aligners..

- Bfast
- BioScope
- Bowtie
- BWA
- CLC bio
- CloudBurst
- Eland/Eland2
- GenomeMapper
- GnuMap
- Karma
- MAQ
- MOM
- Mosaik
- MrFAST/MrsFAST
- NovoAlign
- PASS
- PerM
- Phaster
- RazerS
- RMAP
- SSAHA2
- Segemehl
- SeqMap
- SHRiMP
- Slider/SliderII
- SOAP/SOAP2
- Srprism
- Stampy
- vmatch
- ZOOM
- .....

Subread
STAR

The most efficient aligners are using the **Burrows-Wheeler Transform** with suffix arrays

There are many SNP callers ...

- GATK

UnifiedGenotyper

HaplotypeCaller

- SAMtools/Mpileup/BCFtools/tabix
- FreeBayes
- VarScan - http://varscan.sourceforge.net
- soapSNP – http://soap.genomics.org.cn/soapsnp.html

Common approach is to use 2 to 3 variant callers

Problem is there is little concordance on the variants identified

This has led to a proliferation of Venn diagrams in the literature

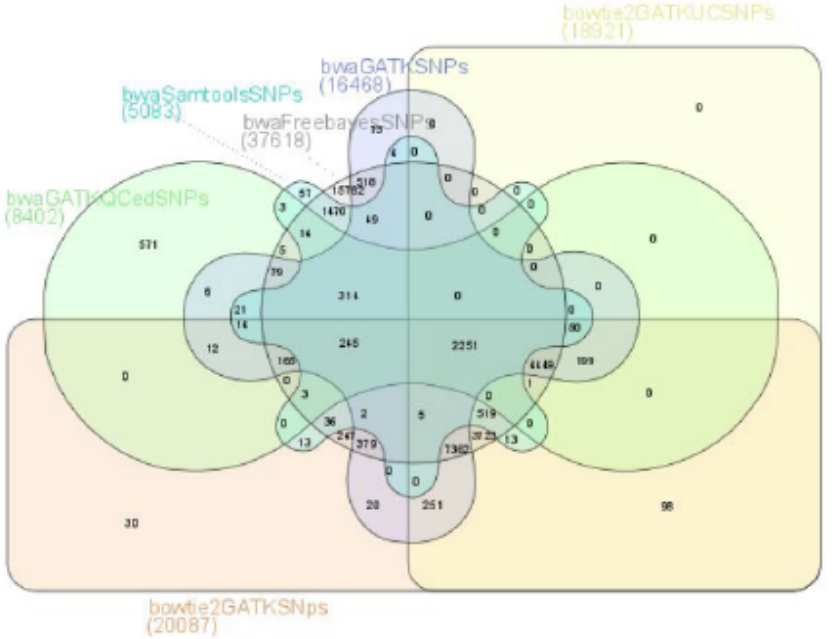


Figure 9. SNP overlap between pipelines (bwaGATK, bwaGATKQCed, bowtie2GATK, bowtie2GATKUG, bwaSamtools, bwaFreebayes).

short introduction in bio-informatics

- Detecting SNPs using 'whole genome resequencing'
  - Current tools
  - File/data formats
- Using 'reduced-representation' sequencing libraries (GBS, RAD-Seq , ...) of multiplexed samples
  - GBS with a reference genome
  - GBS w/o a reference genome : UNEAK and STACKS pipelines
- File/data formats for handling SNPs data for subsequent analyses
- quick view : principle of *de novo* sequencing
- What can I do with this genetic data ?

Summary - Format remind

Reads	FASTQ
↓	
Alignments	SAM/BAM
↓	
Variations	VCF**/BCF** **HAPMAP** **PED/BED**
Genome	FASTA
Genome annotations	GFF/BED

fastq format

- Standard for storing outputs of HTS
- A text-based format for storing a read and its corresponding quality scores
- 1 read <-> 4 lines

```
@HWI-ST218:596:C90JYANXX:8:1101:1293:2188 1:N:0:ATTCAGAATAATCTTA
NCTAAGTGTTAGGGGGTTTCCGCCCCTTAGTGCTGCAGCTAACGCATTAAGCACTCCGCCTGGGGAGTACGGTCGCAAGACTGAAAA
+
#<3?BFGGGGGGGEGGGGGGGEGGGGGG@F1FGGGGGGDDGG1FB</9FE=EGGGGGGGG>GGGGBGGGGG<<C/BDGGGGGGC=GGG
```

1. Begins with '@' character and is followed by a sequence identifier
2. The raw sequence
3. Begins with a '+' character and is optionally followed by the same sequence identifier
4. Encodes the quality values for th read , contains the same number of symbols as letters in the read

fastq format

- Sequence identifier

```
@EAS139:136:FC706VJ:2:2104:15343:197393 1:Y:18:ATCACG
```

1. Begins with '@' character and is followed by a sequence identifier

EAS139	the unique instrument name
136	the run id
FC706VJ	the flowcell id
2	flowcell lane
2104	tile number within the flowcell lane
15343	'x'-coordinate of the cluster within the tile
197393	'y'-coordinate of the cluster within the tile
1	the member of a pair, 1 or 2 *(paired-end or mate-pair reads only)*
Y	Y if the read is filtered, N otherwise
18	0 when none of the control bits are on, otherwise it is an even number
ATCACG	index sequence

fastq format

```
 SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....................................................
 ..........................XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX......................
 ..............................IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII......................
 ................................JJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ......................
 LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL.....................................................
 !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
 |                          |    |       |                              |                     |
33                         59   64      73                            104                   126
 0.........................26...31.......40
                           -5....0........9.............................40
                                0........9.............................40
                                   3.....9.............................41
 0.2......................26...31........41

S - Sanger        Phred+33,  raw reads typically (0, 40)
X - Solexa        Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64,  raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64,  raw reads typically (3, 41)
    with 0=unused, 1=unused, 2=Read Segment Quality Control Indicator (bold)
    (Note: See discussion above).
L - Illumina 1.8+ Phred+33,  raw reads typically (0, 41)
```

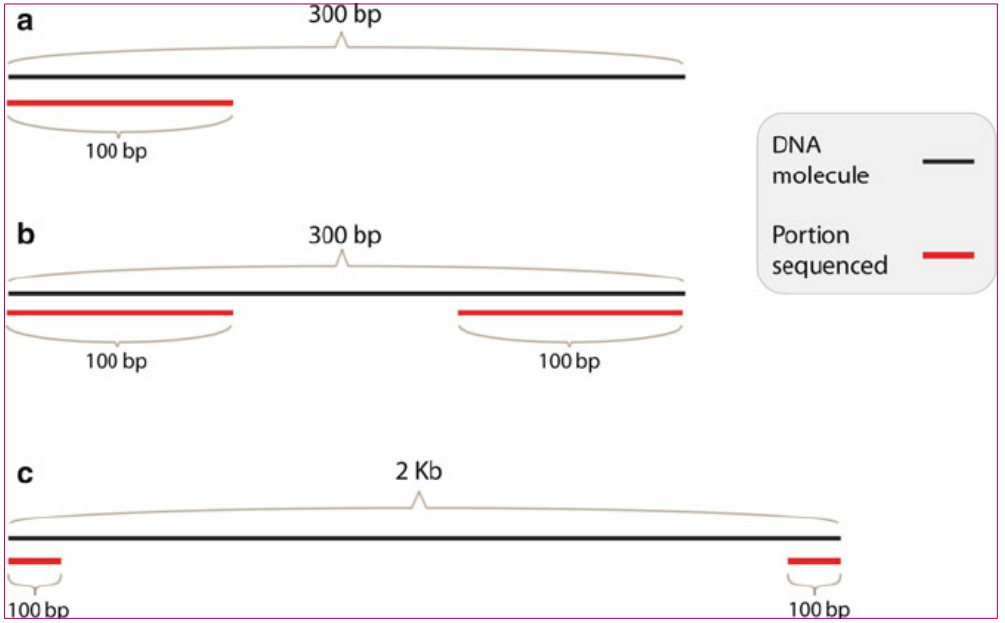
$$Q = -10 \log_{10} P \Leftrightarrow P = 10^{\frac{-Q}{10}}$$

For raw reads, scores depend on technology and the base caller used. Will typically be up to 41 for recent Illumina chemistry.

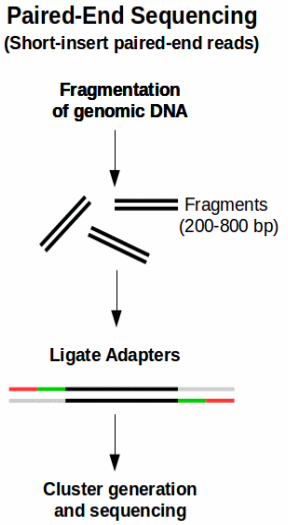
Example : ASCII code = J, probability of ***incorrect*** calling ?
J = 74 , offset = 33, so Q = 41 thus P = 10^(-41/10) = 7.943282E-05

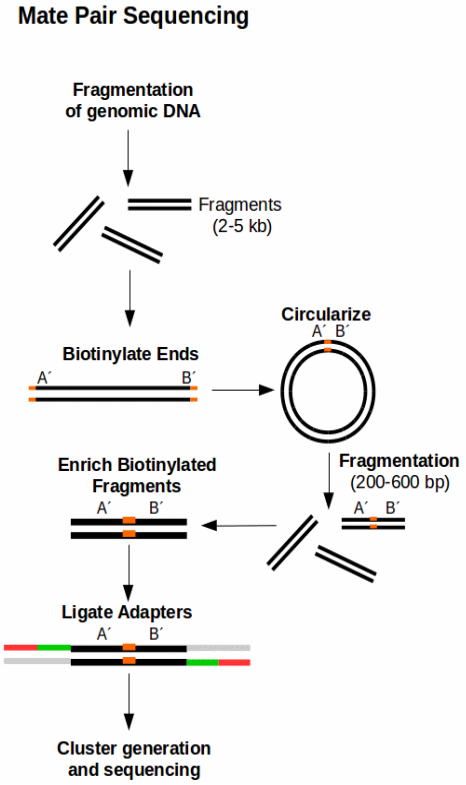
https://en.wikipedia.org/wiki/FASTQ_format

Some vocabulary for reads :



- (a) Single-end read
- (b) Paired-end reads
- (c) Mate-pair reads





Sequence Alignment/Map (SAM) format

- Data sharing was a major issue with the Human 1000 genomes
- SAM format capture all of the critical information about NGS data in a single indexed and compressed file
- Generic alignment format
- Supports short and long reads
- Flexible in style, compact in size, efficient in random access

Website : http://samtools.sourceforge.net/

Binary compressed by BGZF library and indexed formats : BAM and .bai

BIOINFORMATICS APPLICATIONS NOTE
Vol. 25 no. 16 2009, pages 2078–2079
doi:10.1093/bioinformatics/btp352

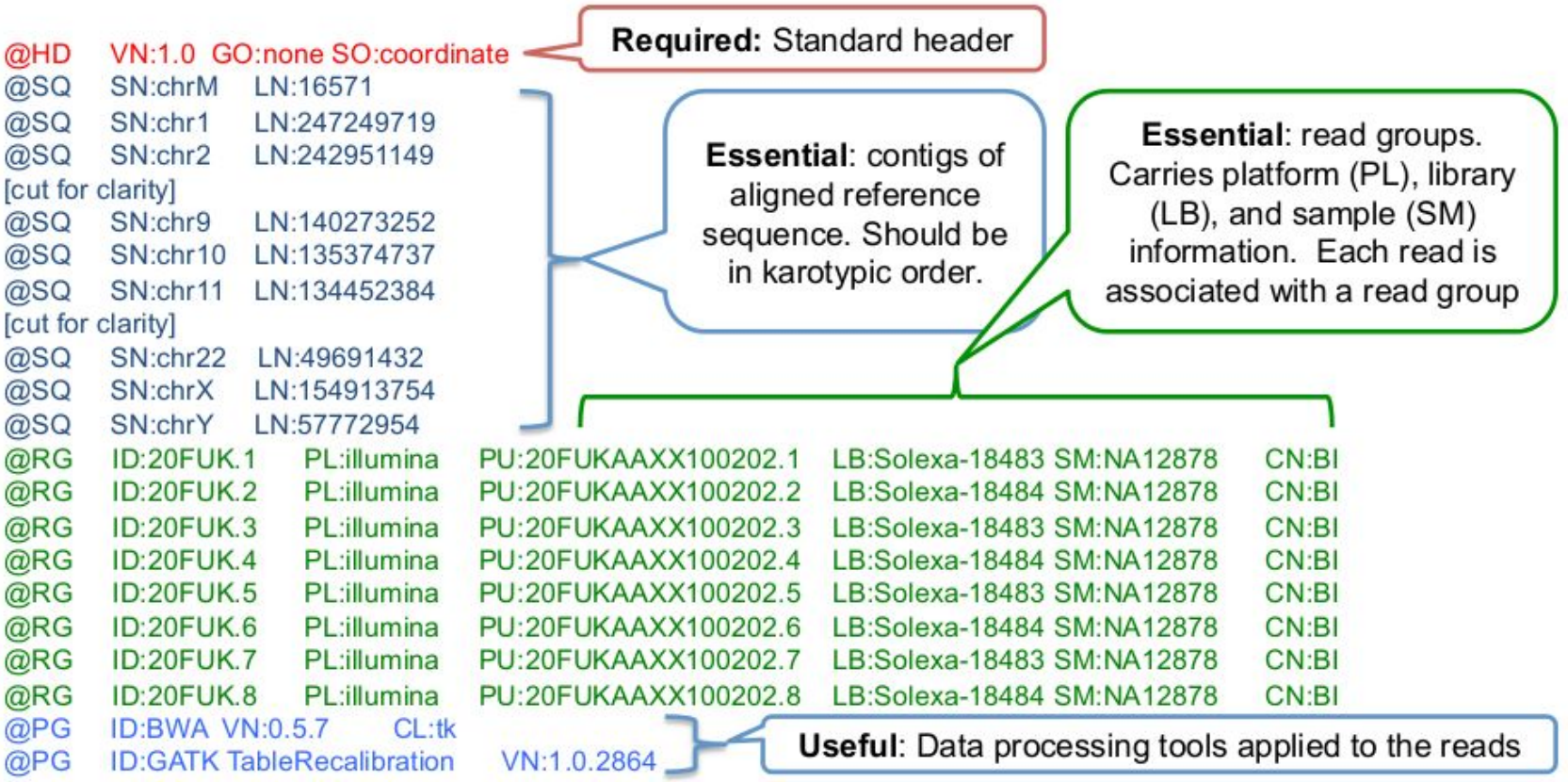
Sequence analysis

The Sequence Alignment/Map format and SAMtools

Heng Li[1,†], Bob Handsaker[2,†], Alec Wysoker[2], Tim Fennell[2], Jue Ruan[3], Nils Homer[4], Gabor Marth[5], Goncalo Abecasis[6], Richard Durbin[1,*] and 1000 Genome Project Data Processing Subgroup[7]

Sequence Alignment/Map (SAM) format

- Header file : generic information



Sequence Alignment/Map (SAM) format

- Body file (alignment description)
  - 11 mandatory fields
  - Variable number of optional fields
  - Tab delimited fields

Col	Field	Description
1	QNAME	Query template/pair NAME
2	FLAG	bitwise FLAG
3	RNAME	Reference sequence NAME
4	POS	1-based leftmost POSition/coordinate of clipped sequence
5	MAPQ	MAPping Quality (Phred-scaled)
6	CIAGR	extended CIGAR string
7	MRNM	Mate Reference sequence NaMe ('=' if same as RNAME)
8	MPOS	1-based Mate POSistion
9	TLEN	inferred Template LENgth (insert size)
10	SEQ	query SEQuence on the same strand as the reference
11	QUAL	query QUALity (ASCII-33 gives the Phred base quality)
12+	OPT	variable OPTional fields in the format TAG:VTYPE:VALUE

Sequence Alignment / Map (SAM) format

```
@SQ SN:chr1 LN:45
r001   99   chr1   7   30   8M2I4M1D3M =   37 39   TTAGATAAAGGATACTG *
r002   0    chr1   9   30   3S6M1P1I4M *   0  0    AAAAGATAAGGATA    *
r003   0    chr1   9   30   5S6M       *   0  0    GCCTAAGCTAA       * SA:Z:ref,29,-,6H5M,17,0;
r004   0    chr1   16  30   6M14N5M    *   0  0    ATAGCTTCAGC       *
r003   2064 chr1   29  17   6H5M       *   0  0    TAGGC             * SA:Z:ref,9,+,5S6M,30,1;
r001   147  chr1   37  30   9M         =   7  -39  CAGCGGCAT          * NM:i:1
```

(1) (2) (3) (4) (5) (6) (7) (8) (9) (10) (11) (12)

Body of SAM file is a tab-delimited format with 11 mandatory fields.

field	Value (read1)	signification
(1)	r001	read name
(2)	99	bitwise Flag with mapping information http://broadinstitute.github.io/picard/explain-flags.html.
(3)	chr1	ref. seq. onto which the read is mapped. '*' if the read is not mapped
(4)	7	Most 5' position of read onto the reference sequence
(5)	30	mapping quality : $-10.\log_{10}(p)$, with p probability of incorrect mapping

Sequence Alignment/Map (SAM) format

```
@SQ SN:chr1 LN:45
r001   99   chr1   7   30   8M2I4M1D3M =   37 39   TTAGATAAAGGATACTG *
r002   0    chr1   9   30   3S6M1P1I4M *   0  0    AAAAGATAAGGATA    *
r003   0    chr1   9   30   5S6M       *   0  0    GCCTAAGCTAA       * SA:Z:ref,29,-,6H5M,17,0;
r004   0    chr1   16  30   6M14N5M    *   0  0    ATAGCTTCAGC       *
r003   2064 chr1   29  17   6H5M       *   0  0    TAGGC             * SA:Z:ref,9,+,5S6M,30,1;
r001   147  chr1   37  30   9M         =   7  -39  CAGCGGCAT          * NM:i:1
```

(1) (2) (3) (4) (5) (6) (7) (8) (9) (10) (11) (12)

field	Value (read1)	signification
(6)	8M2I4M1D3M	CIGAR of the read (see later)
(7)	=	if paired-end: '=' if same chromosome, '*' if single-end read
(8)	37	if paired-end, most 5' position of the pair ; 0 otherwise (no mapping, single-end read)
(9)	39	if paired-end, length of read i.e. difference between the most 5' position of 5' fragment and most 3' position of 3' fragment'
(10)	TTAGATAAA GGATACTG	Sequence of frament
(11)	*	Qualities (as FASTQ), '*' if field not filled

CIGAR for SAM format

```
Ref:  GCATTCAGATGCAGTACGC
Read:   ccTCAG--GCATTAgtg
POS   CIGAR
5     2S4M2D6M3S
```

op	Description
M	Alignment match (can be a sequence match or mismatch)
I	Insertion to the reference
D	Deletion from the reference
N	Skipped region from the reference
S	Soft clip on the read (clipped sequence present in <seq>)
H	Hard clip on the read (clipped sequence NOT present in <seq>)
P	Padding (silent deletion from the padded reference sequence)

* soft-clipped: bases in 5' and 3' of the read are NOT part of the alignment.

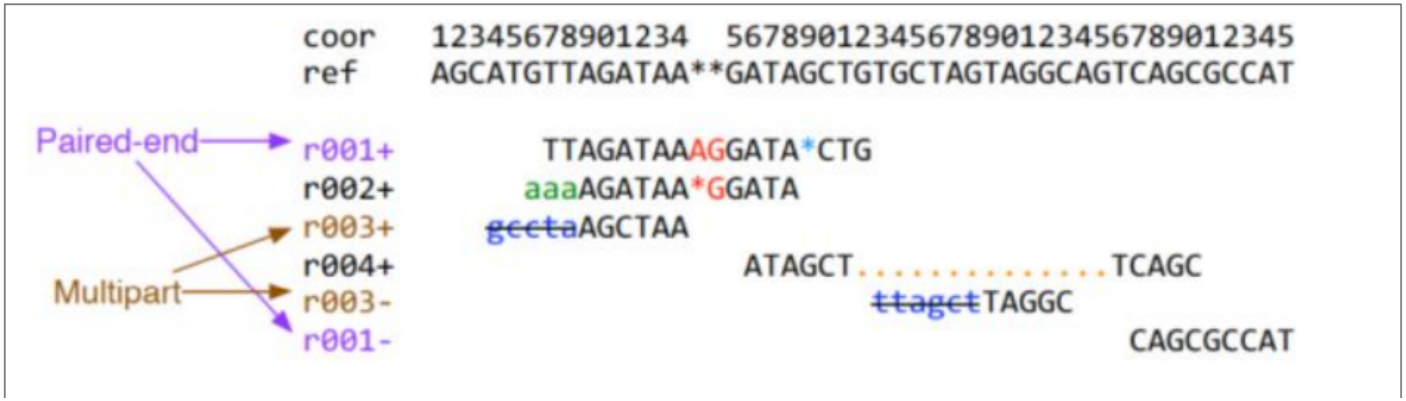
* hard-clipped: bases in 5' and 3' of the read are NOT part of the alignment AND those bases have been removed from the read sequence in the BAM file

CIGAR for SAM format

```
@SQ SN:chr1 LN:45
r001   99   chr1   7   30   8M2I4M1D3M =   37 39   TTAGATAAAGGATACTG *
r002   0    chr1   9   30   3S6M1P1I4M *   0  0    AAAAGATAAGGATA    *
r003   0    chr1   9   30   5S6M       *   0  0    GCCTAAGCTAA       * SA:Z:ref,29,-,6H5M,17,0;
r004   0    chr1   16  30   6M14N5M    *   0  0    ATAGCTTCAGC       *
r003   2064 chr1   29  17   6H5M       *   0  0    TAGGC             * SA:Z:ref,9,+,5S6M,30,1;
r001   147  chr1   37  30   9M         =   7  -39  CAGCGGCAT          * NM:i:1
```

(1) (2) (3) (4) (5) (6) (7) (8) (9) (10) (11) (12)

M: match/mismatch	I: insertion	D: deletion
S: softclip	H: hardclip	P: padding
N: skipp		



The bitwise Flag with mapping information

#	Decimal	Description of read
1	1	Read paired
2	2	Read mapped in proper pair
3	4	Read unmapped
4	8	Mate unmapped
5	16	Read reverse strand
6	32	Mate reverse strand
7	64	First in pair
8	128	Second in pair
9	256	Not primary alignment
10	512	Read fails platform/vendor quality checks
11	1024	Read is PCR or optical duplicate
12	2048	Supplementary alignment

Decimal	Description of second read
1	Read paired
2	Read mapped in proper pair
4	Read unmapped
8	Mate unmapped
16	Read reverse strand
32	Mate reverse strand
64	First in pair
128	Second in pair
256	Not primary alignment
512	Read fails platform/vendor quality checks
1024	Read is PCR or optical duplicate
2048	Supplementary alignment

Common flags*

One of the reads is unmapped:
73, 133, 89, 121, 165, 181, 101, 117, 153, 185, 69, 137

Both reads are unmapped:
77, 141

Mapped within the insert size and in correct orientation:
99, 147, 83, 163

Mapped within the insert size but in wrong orientation:
67, 131, 115, 179

Mapped uniquely, but with wrong insert size:
81, 161, 97, 145, 65, 129, 113, 177

* Collected from here

mapped in correct orientation and within insert size

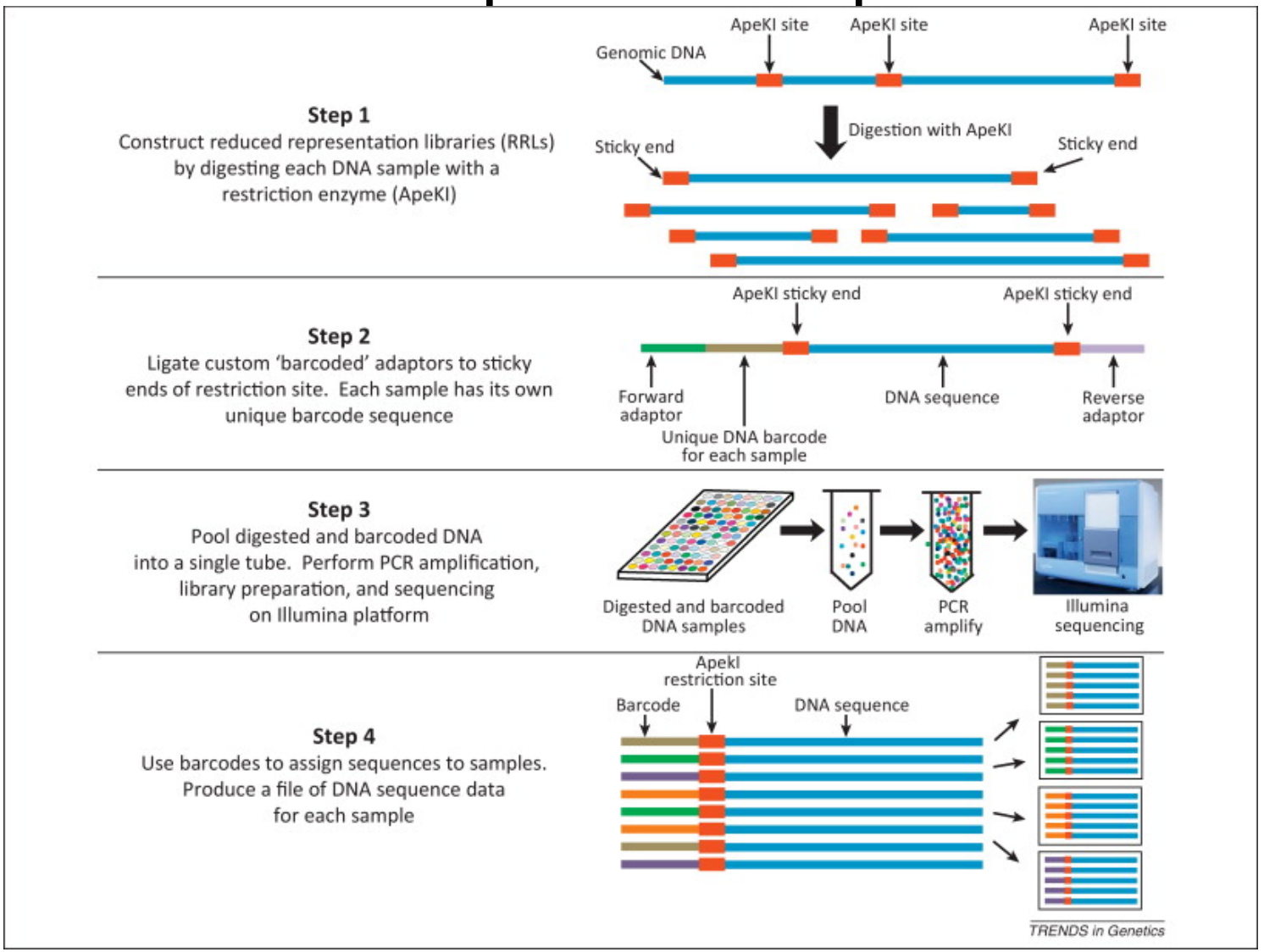
99	1+2+32+64	1	map	+	map	-
147	0+1+2+16+128	2	map	-	map	+

83	1+2+16+64	1	map	-	map	+
163	1+2+32+128	2	map	+	map	-

short introduction in bio-informatics

- Detecting SNPs using 'whole genome resequencing'
  - Current tools
  - File/data formats
- Using 'reduced-representation' sequencing libraries (GBS, RAD-Seq , ...) of multiplexed samples
  - GBS with a reference genome
  - GBS w/o a reference genome : UNEAK and STACKS pipelines
- File/data formats for handling SNPs data for subsequent analyses
- quick view : principle of *de novo* sequencing
- What can I do with this genetic data ?

The basis of reduced-representation sequencing of multiplexed samples



Trends in Genetics (2013) 29, 190-196

reduced-representation sequencing of multiplexed samples

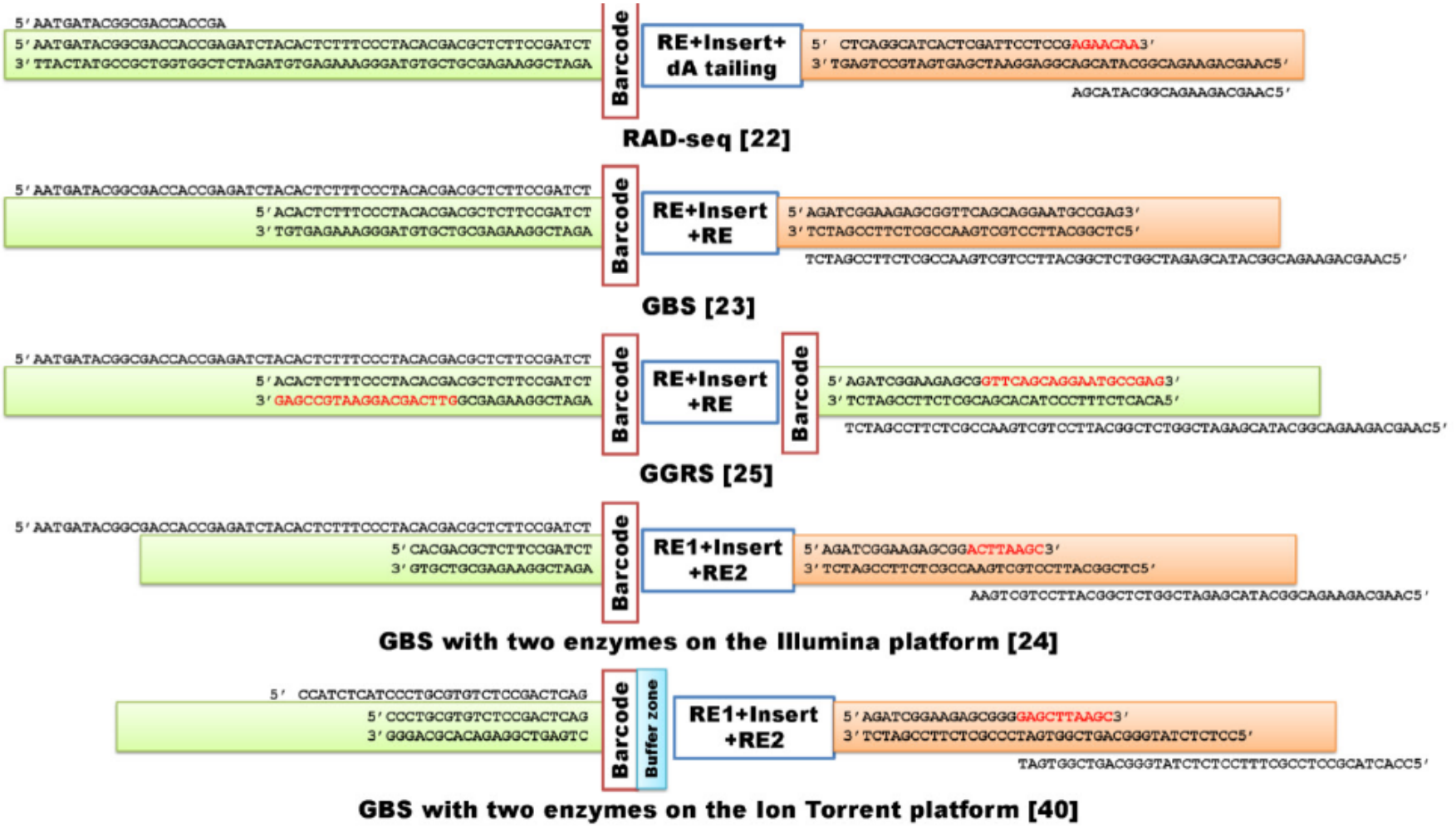


Figure 2. Examples of GWSS NGS library construction methods used to discover genome-wide representative variants, mainly SNPs, among individuals. RE indicates the use of a single restriction enzyme in the assay, while RE1 and RE2 indicate the use of two enzymes for digestion of genomic DNA. The buffer zone contains additional nucleotides to tighten the restriction site ligation. The first four examples are designed for assessment with Illumina analyzers, while the last example is a library built to meet Ion Torrent platform requirements.

Jiang, Z. et al. (2016) International Journal of Biological Sciences 12: 100–108.

Detecting SNPs with Genotyping By Sequencing



1. Quantify Genomic DNA

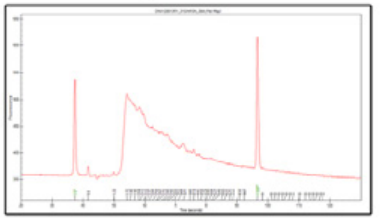


2. Normalize
3. Add Restriction Enzymes
4. Add Adapters and Ligase

5. Pool Samples



6. PCR
7. Clean and Size Library



8. Sequence



FASTQ File

Parse by Barcode

Reference

Align to Reference Genome

Filter SNPs

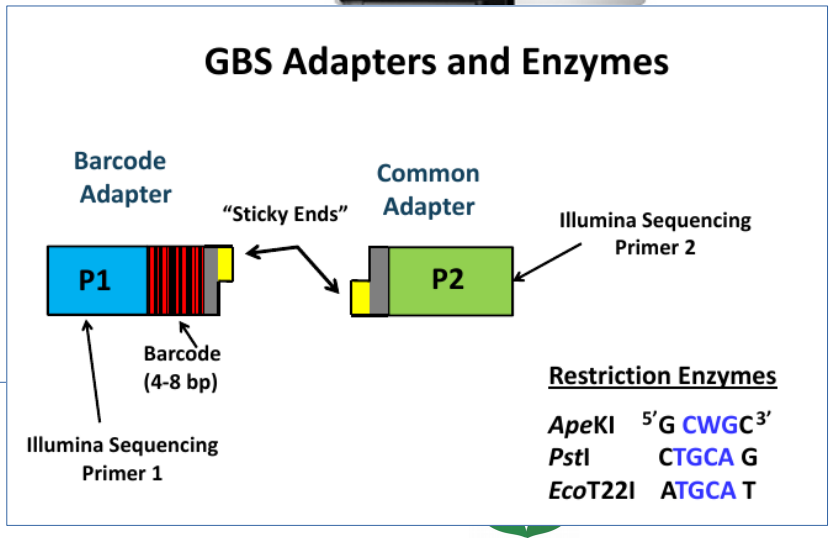
HAP Map

No Reference

Align Tags (Internal)

Filter SNPs

"HAP Map"



Poland et al. (2012) The Plant Genome 5: 92–102.

GBS with reference genome :

TASSEL-GBS v1& v2 – *Glaubitz et al. (2014) PLoS ONE 9: e90346.*

- TASSEL-GBS pipelines are implemented in Java programming language. TASSEL-GBS v1 (TASSEL 3.0) and TASSEL-GBS v2 (TASSEL 5.0). Both pipelines require that all reads be trimmed to an identical length (64 bp in v1, up to 92 bp in v2)

- Stacks – *Catchen et al. (2013) Molecular Ecology 22: 3124–3140.*
  - The raw input data to Stacks are sequenced DNA fragments from any restriction enzyme–based GBS protocol. Stacks can handle raw sequencing data to identify loci de novo or via alignment against a reference genome
- IGST – *Sonah et al. (2013) PLoS One 8: e54603.*
  - A pipeline implemented in Perl programming language developed for the processing of Illumina sequence read data.
- Fast-GBS – *Torkamaneh et al. (2017) BMC Bioinformatics 18: 5.*
  - developed by integrating public packages with internally developed tools

- Others ….

Torkamaneh et al. (2016) PLoS One 11: e0161333.

Example of Fast-GBS

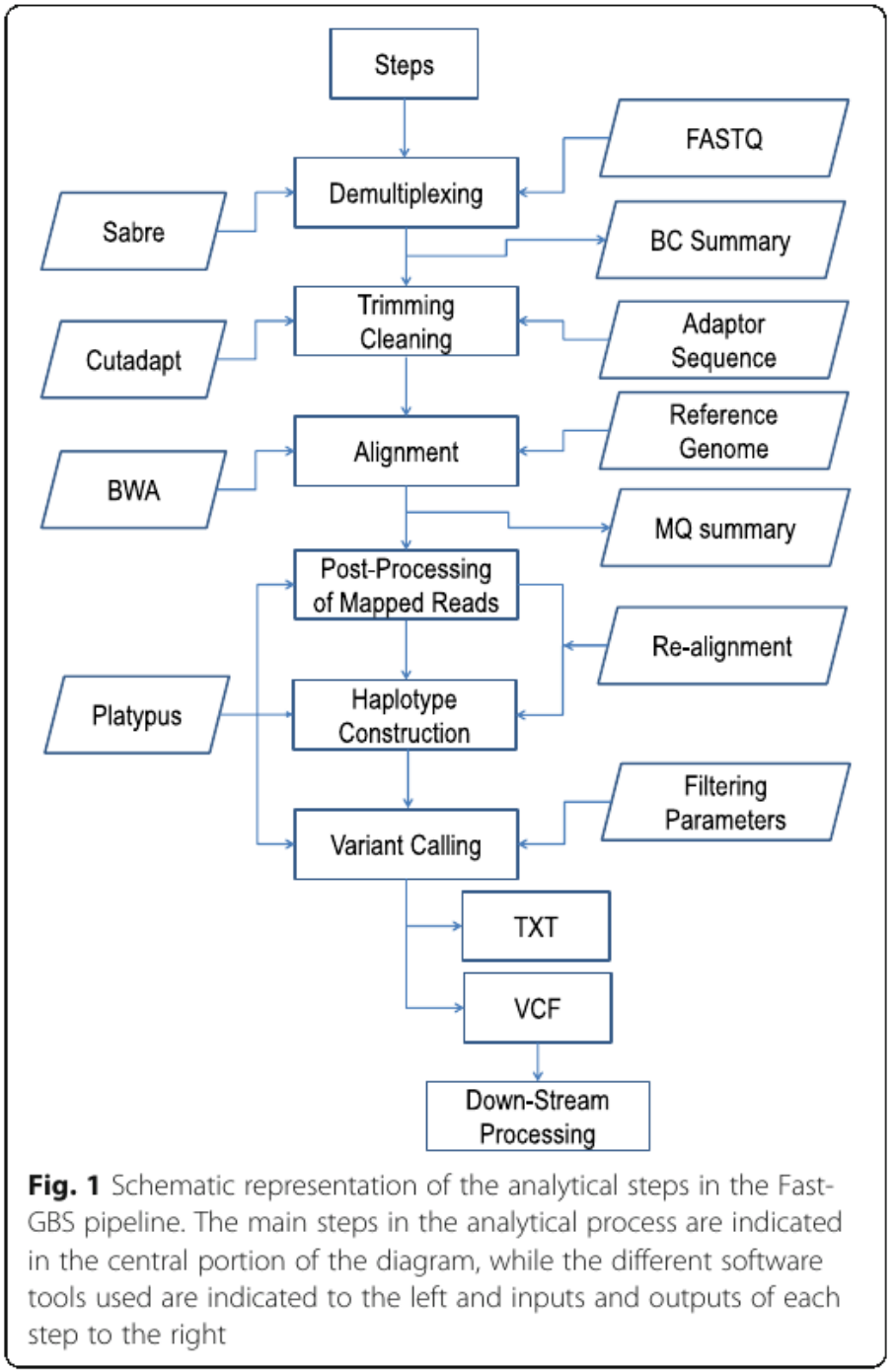


Fig. 1 Schematic representation of the analytical steps in the Fast-GBS pipeline. The main steps in the analytical process are indicated in the central portion of the diagram, while the different software tools used are indicated to the left and inputs and outputs of each step to the right

Torkamaneh et al. (2017) BMC Bioinformatics 18: 5.

GBS w/o reference genome : UNEAK pipeline

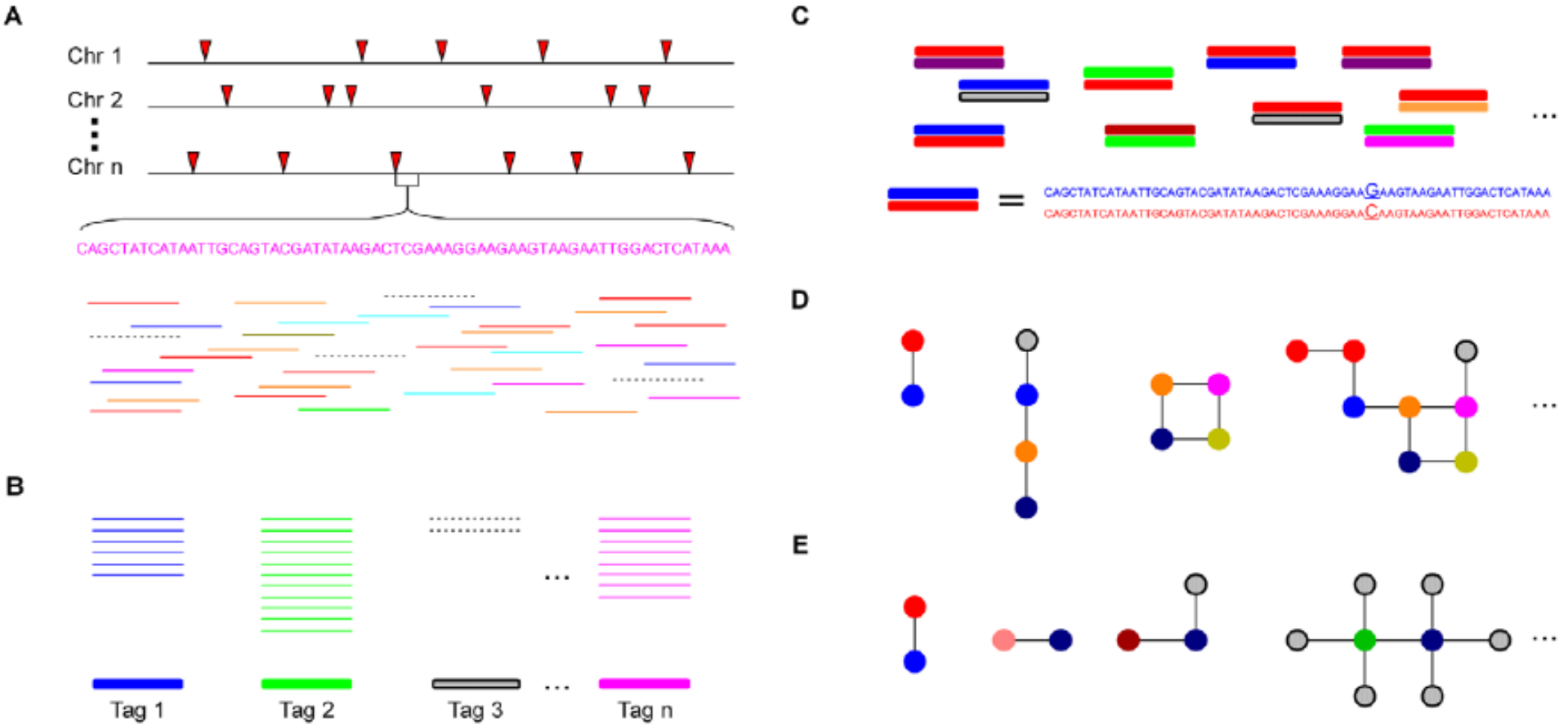


Figure 1. The analytical framework of UNEAK. (A) Multiple DNA samples are digested and sequenced using GBS (red arrows represent cut sites). The inputs of UNEAK are Illumina Qseq or Fastq files. All of the reads are computationally trimmed to 64 bp. The solid colored lines represent error-free ("real") reads, while the dashed lines are reads containing one or more sequencing errors. (B) Identical reads are classified as a tag. The colored bars are real tags, whereas the shaded bar is a rarer error tag. (C) Pairwise alignment is performed to find tag pairs differing by only a single bp mismatch. (D) Topology of tag networks. The colored circles are real tags. The shaded circles are error tags. Lines ("edges") are drawn only between tags that differ by a single bp mismatch. (E) Only reciprocal, real tag pairs are retained as SNPs.
doi:10.1371/journal.pgen.1003215.g001

GBS w/o reference genome : Stacks pipeline

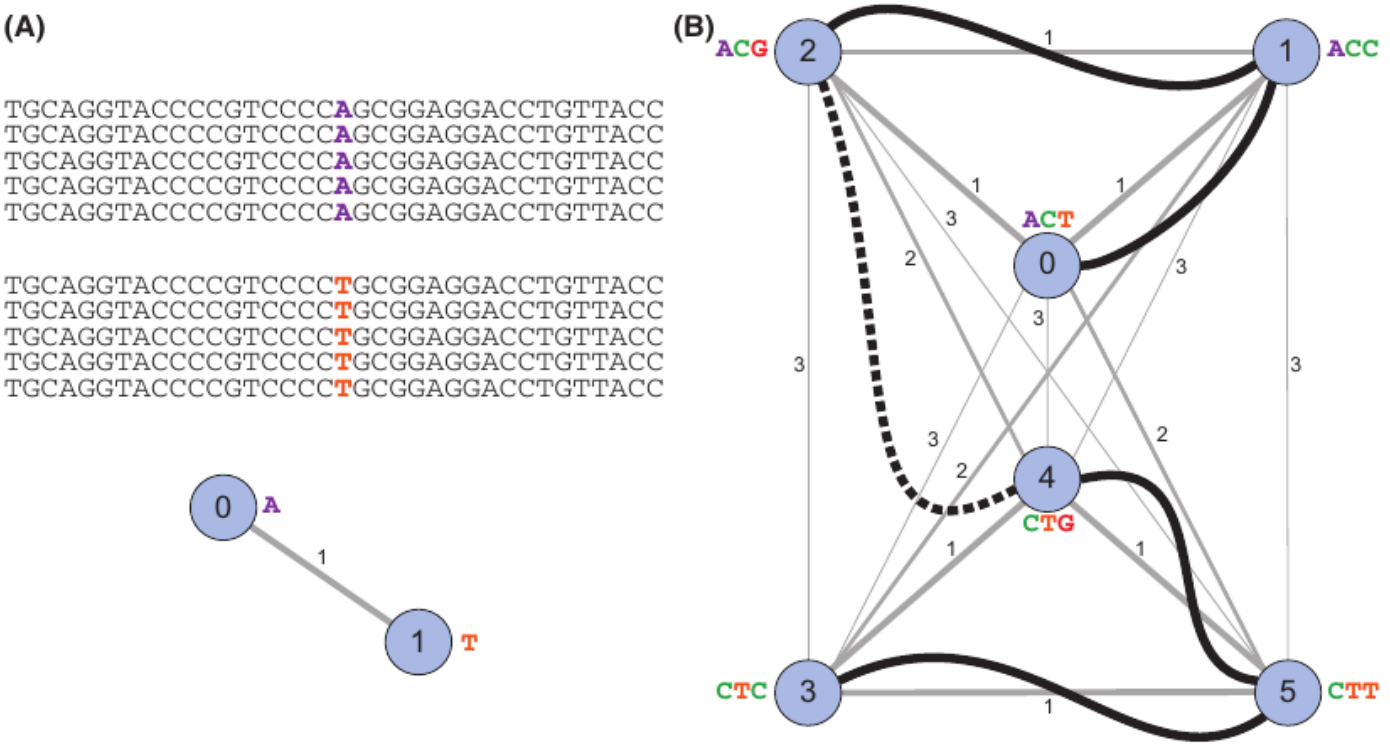


Fig. 2 The ustacks deleveraging algorithm. (A) The simplest polymorphic locus is defined by a single SNP (A/T), and as the remainder of the locus is identical in both alleles, we can refer to the entire locus by the A/T haplotypes. A locus can be visualized as an undirected graph, with each allele or stack as a node, and with the nodes connected by an edge weighted according to the nucleotide distance between them. (B) This locus with three detected polymorphisms comprises six distinct stacks, which is not biologically possible and must be the result of either erroneous stacks or collapsed, repetitive loci. The deleveraging algorithm calculates a minimum-spanning tree from the locus (thick, black lines), calculates the minimum distance between any two nodes and breaks edges (separating loci) whenever they are connected by edges larger than the minimum edge. The result in this case is two loci, the first built from stacks 0, 1 and 2, and the second built from stacks 3, 4 and 5.

Catchen et al. (2013) Molecular Ecology 22: 3124–3140.

http://catchenlab.life.illinois.edu/stacks/manual/

GBS pipelines to produce a catalogue of SNPs starting from reads in a FASTQ file.

Torkamaneh et al. (2016) PLoS One 11: e0161333.

Step/Pipeline	***TASSEL-GBS v1***	***Stacks***	***IGST***	***TASSEL-GBS v2***	***Fast-GBS***
Demultiplexer	FasqToTagCount	Stacks	BarcodeSplitter	GBSSeqToTagDB	Sabre
Trimmer	FasqToTagCount	Cutadapt	FASTX-Toolkit	GBSSeqToTagDB	Cutadapt
Read/Tag	Tag	Read	Read	Tag	Read
Minimum/Maximum Read/Tag length (bp)	32-64	>50	>50	20-92	>50
Aligner	BWA-aln	BWA-mem	BWA-aln	BWA-aln	BWA-mem
Post-processing of mapped reads	No	No	No	No	Yes
Variant caller	GBSTagToSNP	pstacks	SAMtools	DiscoverySNPCaller v2	Platypus
Single/Multi-sample variant calling	Single	Multi	Single	Single	Multi

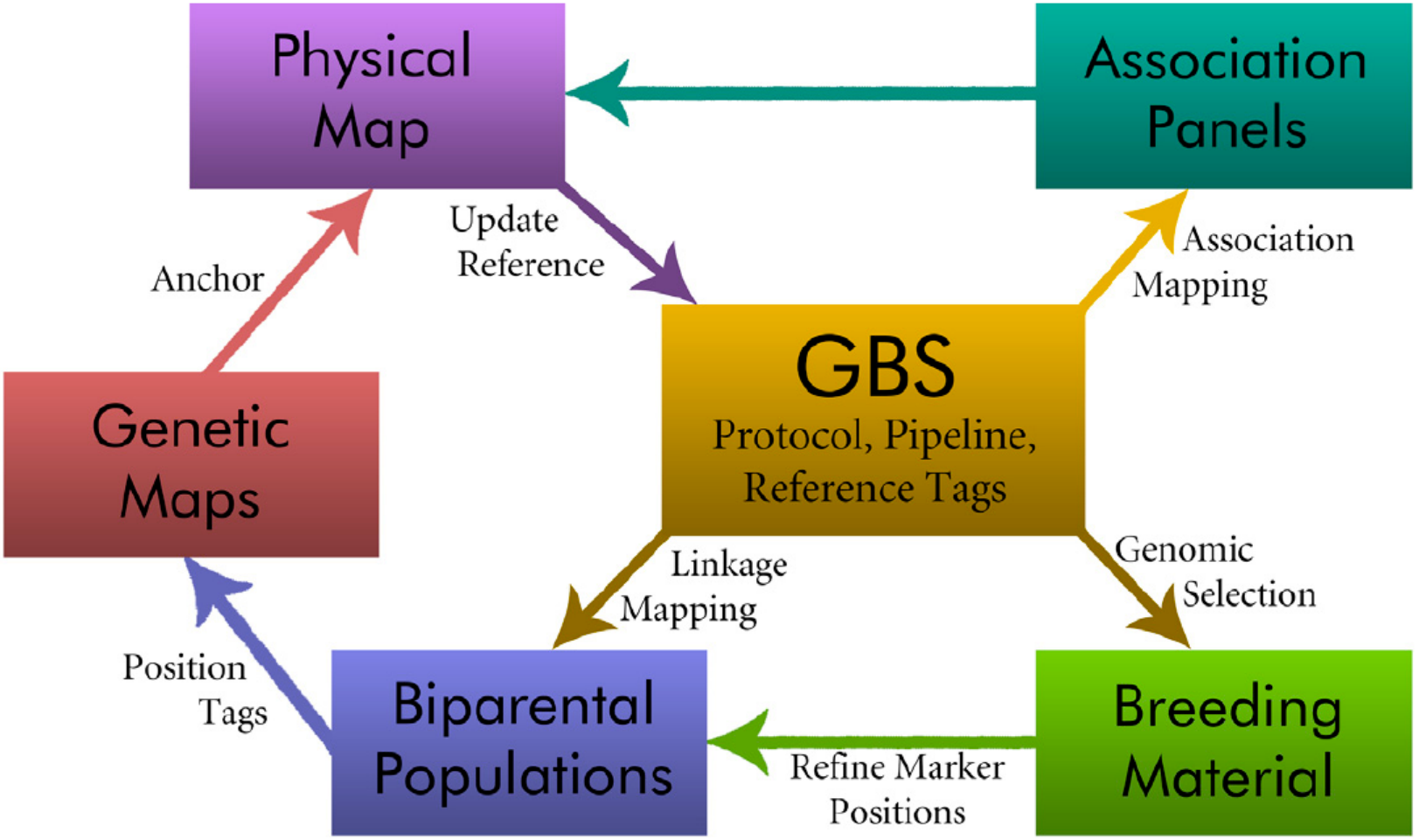


Figure 3. Integration of genotyping-by-sequencing (GBS) in the context of plant breeding and genomics for a species without a completed reference genome.

Poland et al. (2012) The Plant Genome 5: 92–102.

short introduction in bio-informatics

- Detecting SNPs using 'whole genome resequencing'
  - Current tools
  - File/data formats
- Using 'reduced-representation' sequencing libraries (GBS, RAD-Seq , ...) of multiplexed samples
  - GBS with a reference genome
  - GBS w/o a reference genome : UNEAK and STACKS pipelines
- File/data formats for handling SNPs data for subsequent analyses
- quick view : principle of *de novo* sequencing
- What can I do with this genetic data ?

Summary - Format remind

Reads	FASTQ
↓	
Alignments	SAM/BAM
↓	
Variations	VCF**/BCF** **HAPMAP** **PED/BED**
Genome	FASTA
Genome annotations	GFF/BED

The Variant calling Format VCF

- http://vcftools.sourceforge.net/specs.html
- Tab-delimited file
- Basic documentation inside
- Header lines start with ## or #
- Give informations of data/tools/parameters used
-
- Variant lines represent a position on the genome
  - Exemaple : 850,000 SNPs onto 300 accessions :

    a file with (at least) 850,000 lines and (at least 300 columns)

The VCF header

VCF version

```
##fileformat=VCFv4.2
##FILTER=<ID=PASS,Description="All filters passed">
##FILTER=<ID=DP_GT_5000,Description="Combined sample read depth exceeded 5000">
##FILTER=<ID=NON_SNP,Description="Variant was not a single base pair substitution">
##FILTER=<ID=MULTI_ALT_ALLELE,Description="Variant had more than one alternate allele">
##FILTER=<ID=LowQual,Description="Low quality">
##FORMAT=<ID=AD,Number=.,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="Normalized, Phred-scaled likelihoods for genotypes as defined in the VCF specification">
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each ALT allele, in the same order as listed">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##INFO=<ID=BaseQRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt Vs. Ref base qualities">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth; some reads may have been filtered">
##INFO=<ID=DS,Number=0,Type=Flag,Description="Were any of the samples downsampled?">
##INFO=<ID=Dels,Number=1,Type=Float,Description="Fraction of Reads Containing Spanning Deletions">
##INFO=<ID=FS,Number=1,Type=Float,Description="Phred-scaled p-value using Fisher's exact test to detect strand bias">
##INFO=<ID=HaplotypeScore,Number=1,Type=Float,Description="Consistency of the site with at most two segregating haplotypes">
##INFO=<ID=InbreedingCoeff,Number=1,Type=Float,Description="Inbreeding coefficient as estimated from the genotype likelihoods per-sample
when compared against the Hardy-Weinberg expectation">
##INFO=<ID=MLEAC,Number=A,Type=Integer,Description="Maximum likelihood expectation (MLE) for the allele counts (not necessarily the same
as the AC), for each ALT allele, in the same order as listed">
##INFO=<ID=MLEAF,Number=A,Type=Float,Description="Maximum likelihood expectation (MLE) for the allele frequency (not necessarily the same
as the AF), for each ALT allele, in the same order as listed">
##INFO=<ID=MQ,Number=1,Type=Float,Description="RMS Mapping Quality">
##INFO=<ID=MQ0,Number=1,Type=Integer,Description="Total Mapping Quality Zero Reads">
##INFO=<ID=MQRankSum,Number=1,Type=Float,Description="Z-score From Wilcoxon rank sum test of Alt vs. Ref read mapping qualities">
##INFO=<ID=QD,Number=1,Type=Float,Description="Variant Confidence/Quality by Depth">
##INFO=<ID=RPA,Number=.,Type=Integer,Description="Number of times tandem repeat unit is repeated, for each allele (including reference)">
##INFO=<ID=RU,Number=1,Type=String,Description="Tandem repeat unit (bases)">
##INFO=<ID=ReadPosRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt vs. Ref read position bias">
##INFO=<ID=STR,Number=0,Type=Flag,Description="Variant is a short tandem repeat">
```

The VCF header

Filtering options used to generate this file

```
##fileformat=VCFv4.2
##FILTER=<ID=PASS,Description="All filters passed">
##FILTER=<ID=DP_GT_5000,Description="Combined sample read depth exceeded 5000">
##FILTER=<ID=NON_SNP,Description="Variant was not a single base pair substitution">
##FILTER=<ID=MULTI_ALT_ALLELE,Description="Variant had more than one alternate allele">
##FILTER=<ID=LowQual,Description="Low quality">
##FORMAT=<ID=AD,Number=.,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="Normalized, Phred-scaled likelihoods for genotypes as defined in the VCF specification">
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each ALT allele, in the same order as listed">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##INFO=<ID=BaseQRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt Vs. Ref base qualities">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth; some reads may have been filtered">
##INFO=<ID=DS,Number=0,Type=Flag,Description="Were any of the samples downsampled?">
##INFO=<ID=Dels,Number=1,Type=Float,Description="Fraction of Reads Containing Spanning Deletions">
##INFO=<ID=FS,Number=1,Type=Float,Description="Phred-scaled p-value using Fisher's exact test to detect strand bias">
##INFO=<ID=HaplotypeScore,Number=1,Type=Float,Description="Consistency of the site with at most two segregating haplotypes">
##INFO=<ID=InbreedingCoeff,Number=1,Type=Float,Description="Inbreeding coefficient as estimated from the genotype likelihoods per-sample
when compared against the Hardy-Weinberg expectation">
##INFO=<ID=MLEAC,Number=A,Type=Integer,Description="Maximum likelihood expectation (MLE) for the allele counts (not necessarily the same
as the AC), for each ALT allele, in the same order as listed">
##INFO=<ID=MLEAF,Number=A,Type=Float,Description="Maximum likelihood expectation (MLE) for the allele frequency (not necessarily the same
as the AF), for each ALT allele, in the same order as listed">
##INFO=<ID=MQ,Number=1,Type=Float,Description="RMS Mapping Quality">
##INFO=<ID=MQ0,Number=1,Type=Integer,Description="Total Mapping Quality Zero Reads">
##INFO=<ID=MQRankSum,Number=1,Type=Float,Description="Z-score From Wilcoxon rank sum test of Alt vs. Ref read mapping qualities">
##INFO=<ID=QD,Number=1,Type=Float,Description="Variant Confidence/Quality by Depth">
##INFO=<ID=RPA,Number=.,Type=Integer,Description="Number of times tandem repeat unit is repeated, for each allele (including reference)">
##INFO=<ID=RU,Number=1,Type=String,Description="Tandem repeat unit (bases)">
##INFO=<ID=ReadPosRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt vs. Ref read position bias">
##INFO=<ID=STR,Number=0,Type=Flag,Description="Variant is a short tandem repeat">
```

The VCF header

```
##fileformat=VCFv4.2
##FILTER=<ID=PASS,Description="All filters passed">
##FILTER=<ID=DP_GT_5000,Description="Combined sample read depth exceeded 5000">
##FILTER=<ID=NON_SNP,Description="Variant was not a single base pair substitution">
##FILTER=<ID=MULTI_ALT_ALLELE,Description="Variant had more than one alternate allele">
##FILTER=<ID=LowQual,Description="Low quality">
##FORMAT=<ID=AD,Number=.,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="Normalized, Phred-scaled likelihoods for genotypes as defined in the VCF specification">
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each ALT allele, in the same order as listed">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##INFO=<ID=BaseQRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt Vs. Ref base qualities">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth; some reads may have been filtered">
##INFO=<ID=DS,Number=0,Type=Flag,Description="Were any of the samples downsampled?">
##INFO=<ID=Dels,Number=1,Type=Float,Description="Fraction of Reads Containing Spanning Deletions">
##INFO=<ID=FS,Number=1,Type=Float,Description="Phred-scaled p-value using Fisher's exact test to detect strand bias">
##INFO=<ID=HaplotypeScore,Number=1,Type=Float,Description="Consistency of the site with at most two segregating haplotypes">
##INFO=<ID=InbreedingCoeff,Number=1,Type=Float,Description="Inbreeding coefficient as estimated from the genotype likelihoods per-sample
when compared against the Hardy-Weinberg expectation">
##INFO=<ID=MLEAC,Number=A,Type=Integer,Description="Maximum likelihood expectation (MLE) for the allele counts (not necessarily the same
as the AC), for each ALT allele, in the same order as listed">
##INFO=<ID=MLEAF,Number=A,Type=Float,Description="Maximum likelihood expectation (MLE) for the allele frequency (not necessarily the same
as the AF), for each ALT allele, in the same order as listed">
##INFO=<ID=MQ,Number=1,Type=Float,Description="RMS Mapping Quality">
##INFO=<ID=MQ0,Number=1,Type=Integer,Description="Total Mapping Quality Zero Reads">
##INFO=<ID=MQRankSum,Number=1,Type=Float,Description="Z-score From Wilcoxon rank sum test of Alt vs. Ref read mapping qualities">
##INFO=<ID=QD,Number=1,Type=Float,Description="Variant Confidence/Quality by Depth">
##INFO=<ID=RPA,Number=.,Type=Integer,Description="Number of times tandem repeat unit is repeated, for each allele (including reference)">
##INFO=<ID=RU,Number=1,Type=String,Description="Tandem repeat unit (bases)">
##INFO=<ID=ReadPosRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt vs. Ref read position bias">
##INFO=<ID=STR,Number=0,Type=Flag,Description="Variant is a short tandem repeat">
```

Genotyping data

The VCF header

```
##fileformat=VCFv4.2
##FILTER=<ID=PASS,Description="All filters passed">
##FILTER=<ID=DP_GT_5000,Description="Combined sample read depth exceeded 5000">
##FILTER=<ID=NON_SNP,Description="Variant was not a single base pair substitution">
##FILTER=<ID=MULTI_ALT_ALLELE,Description="Variant had more than one alternate allele">
##FILTER=<ID=LowQual,Description="Low quality">
##FORMAT=<ID=AD,Number=.,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="Normalized, Phred-scaled likelihoods for genotypes as defined in the VCF specification">
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each ALT allele, in the same order as listed">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##INFO=<ID=BaseQRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt Vs. Ref base qualities">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth; some reads may have been filtered">
##INFO=<ID=DS,Number=0,Type=Flag,Description="Were any of the samples downsampled?">
##INFO=<ID=Dels,Number=1,Type=Float,Description="Fraction of Reads Containing Spanning Deletions">
##INFO=<ID=FS,Number=1,Type=Float,Description="Phred-scaled p-value using Fisher's exact test to detect strand bias">
##INFO=<ID=HaplotypeScore,Number=1,Type=Float,Description="Consistency of the site with at most two segregating haplotypes">
##INFO=<ID=InbreedingCoeff,Number=1,Type=Float,Description="Inbreeding coefficient as estimated from the genotype likelihoods per-sample
when compared against the Hardy-Weinberg expectation">
##INFO=<ID=MLEAC,Number=A,Type=Integer,Description="Maximum likelihood expectation (MLE) for the allele counts (not necessarily the same
as the AC), for each ALT allele, in the same order as listed">
##INFO=<ID=MLEAF,Number=A,Type=Float,Description="Maximum likelihood expectation (MLE) for the allele frequency (not necessarily the same
as the AF), for each ALT allele, in the same order as listed">
##INFO=<ID=MQ,Number=1,Type=Float,Description="RMS Mapping Quality">
##INFO=<ID=MQ0,Number=1,Type=Integer,Description="Total Mapping Quality Zero Reads">
##INFO=<ID=MQRankSum,Number=1,Type=Float,Description="Z-score From Wilcoxon rank sum test of Alt vs. Ref read mapping qualities">
##INFO=<ID=QD,Number=1,Type=Float,Description="Variant Confidence/Quality by Depth">
##INFO=<ID=RPA,Number=.,Type=Integer,Description="Number of times tandem repeat unit is repeated, for each allele (including reference)">
##INFO=<ID=RU,Number=1,Type=String,Description="Tandem repeat unit (bases)">
##INFO=<ID=ReadPosRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt vs. Ref read position bias">
##INFO=<ID=STR,Number=0,Type=Flag,Description="Variant is a short tandem repeat">
```

Additional data

The VCF header

```
##contig=<ID=chr1,length=52991155>
##contig=<ID=chr2,length=45729672>
##contig=<ID=chr3,length=55515152>
##contig=<ID=chr4,length=56582383>
##contig=<ID=chr5,length=43630510>
##contig=<ID=chr6,length=35275713>
##contig=<ID=chr7,length=49172423>
##contig=<ID=chr8,length=45569985>
```

```
##contig=<ID=chr1,length=52991155>_
##contig=<ID=chr2,length=45729672>_
##contig=<ID=chr3,length=55515152>_
##contig=<ID=chr4,length=56582383>_
##contig=<ID=chr5,length=43630510>_
##contig=<ID=chr6,length=35275713>_
##contig=<ID=chr7,length=49172423>_
##contig=<ID=chr8,length=45569985>_
```

Genome information

```
##bcftools_viewCommand=view -O v ./chr7-filtered-set-2014Apr15.TasselQSNPs.bcf
##bcftools_viewCommand=view -O v ./chr7-filtered-set-2014Apr15.TasselQSNPs.bcf_
#CHROM POS    ID     REF    ALT    QUAL   FILTER INFO   FORMAT HM004  HM005  HM006  HM101

chr7   157    chr7_157     T      G      8510.72      PASS
AC=178;AF=0.78;AN=214;BaseQRankSum=0.998;DP=800;Dels=0;FS=25.28;HaplotypeScore=2.187;InbreedingCoeff=0.2401;MLEAC=199;MLEAF=0.754;MQ=37.
62;MQ0=0;MQRankSum=-4.652;QD=13.68;ReadPosRankSum=0.716    GT:AD:DP:GQ:PL      ./.:.:.:.:.  1/1:0,6:6:12:143,12,0
1/1:0,3:3:9:102,9,0 0/0:54,1:54:99:0,132,1529

chr7   261    chr7_261     T      C      4731.37      PASS
AC=176;AF=0.71;AN=212;BaseQRankSum=4.85;DP=749;Dels=0;FS=78.113;HaplotypeScore=1.6073;InbreedingCoeff=0.4641;MLEAC=176;MLEAF=0.638;MQ=37
.66;MQ0=2;MQRankSum=-5.928;QD=20.48;ReadPosRankSum=4.061   GT:AD:DP:GQ:PL      1/1:0,2:2:6:54,6,0  1/1:0,11:11:24:267,24,0
1/1:2,4:6:3:29,3,0  0/0:70,2:70:99:0,195,2207

chr7   553    chr7_553     G      T      10708.8      PASS
AC=277;AF=0.813;AN=310;BaseQRankSum=4.498;DP=1298;Dels=0.01;FS=342.229;HaplotypeScore=2.0697;InbreedingCoeff=0.3771;MLEAC=257;MLEAF=0.70
6;MQ=36.34;MQ0=0;MQRankSum=-12.248;QD=16.65;ReadPosRankSum=6.033  GT:AD:DP:GQ:PL      1/1:0,4:4:9:82,9,0  1/1:0,5:5:12:135,12,0
1/1:0,7:7:15:162,15,0      0/0:118,8:120:99:0,253,3653
```

The VCF header

```
##contig=<ID=chr1,length=52991155>
##contig=<ID=chr2,length=45729672>
##contig=<ID=chr3,length=55515152>
##contig=<ID=chr4,length=56582383>
##contig=<ID=chr5,length=43630510>
##contig=<ID=chr6,length=35275713>
##contig=<ID=chr7,length=49172423>
##contig=<ID=chr8,length=45569985>
```

```
##contig=<ID=chr1,length=52991155>_
##contig=<ID=chr2,length=45729672>_
##contig=<ID=chr3,length=55515152>_
##contig=<ID=chr4,length=56582383>_
##contig=<ID=chr5,length=43630510>_
##contig=<ID=chr6,length=35275713>_
##contig=<ID=chr7,length=49172423>_
##contig=<ID=chr8,length=45569985>_
```

Header line

```
##bcftools_viewCommand=view -O v ./chr7-filtered-set-2014Apr15.TasselQSNPs.bcf
```

```
##bcftools_viewCommand=view -O v ./chr7-filtered-set-2014Apr15.TasselQSNPs.bcf_
```

```
#CHROM POS     ID      REF     ALT     QUAL    FILTER INFO    FORMAT HM004  HM005  HM006  HM101

chr7    157     chr7_157        T       G       8510.72         PASS
AC=178;AF=0.78;AN=214;BaseQRankSum=0.998;DP=800;Dels=0;FS=25.28;HaplotypeScore=2.187;InbreedingCoeff=0.2401;MLEAC=199;MLEAF=0.754;MQ=37.
62;MQ0=0;MQRankSum=-4.652;QD=13.68;ReadPosRankSum=0.716     GT:AD:DP:GQ:PL      ./.:.:.:.:.   1/1:0,6:6:12:143,12,0
1/1:0,3:3:9:102,9,0 0/0:54,1:54:99:0,132,1529

chr7    261     chr7_261        T       C       4731.37         PASS
AC=176;AF=0.71;AN=212;BaseQRankSum=4.85;DP=749;Dels=0;FS=78.113;HaplotypeScore=1.6073;InbreedingCoeff=0.4641;MLEAC=176;MLEAF=0.638;MQ=37
.66;MQ0=2;MQRankSum=-5.928;QD=20.48;ReadPosRankSum=4.061    GT:AD:DP:GQ:PL      1/1:0,2:2:6:54,6,0  1/1:0,11:11:24:267,24,0
1/1:2,4:6:3:29,3,0  0/0:70,2:70:99:0,195,2207

chr7    553     chr7_553        G       T       10708.8         PASS
AC=277;AF=0.813;AN=310;BaseQRankSum=4.498;DP=1298;Dels=0.01;FS=342.229;HaplotypeScore=2.0697;InbreedingCoeff=0.3771;MLEAC=257;MLEAF=0.70
6;MQ=36.34;MQ0=0;MQRankSum=-12.248;QD=16.65;ReadPosRankSum=6.033  GT:AD:DP:GQ:PL      1/1:0,4:4:9:82,9,0  1/1:0,5:5:12:135,12,0
1/1:0,7:7:15:162,15,0      0/0:118,8:120:99:0,253,3653
```

The VCF body : genotype data

```
##contig=<ID=chr1,length=52991155>
##contig=<ID=chr2,length=45729672>
##contig=<ID=chr3,length=55515152>
##contig=<ID=chr4,length=56582383>
##contig=<ID=chr5,length=43630510>
##contig=<ID=chr6,length=35275713>
##contig=<ID=chr7,length=49172423>
##contig=<ID=chr8,length=45569985>
```

```
##contig=<ID=chr1,length=52991155>_
##contig=<ID=chr2,length=45729672>_
##contig=<ID=chr3,length=55515152>_
##contig=<ID=chr4,length=56582383>_
##contig=<ID=chr5,length=43630510>_
##contig=<ID=chr6,length=35275713>_
##contig=<ID=chr7,length=49172423>_
##contig=<ID=chr8,length=45569985>_
```

```
##bcftools_viewCommand=view -O v ./chr7-filtered-set-2014Apr15.TasselQSNPs.bcf
```

```
##bcftools_viewCommand=view -O v ./chr7-filtered-set-2014Apr15.TasselQSNPs.bcf_
```

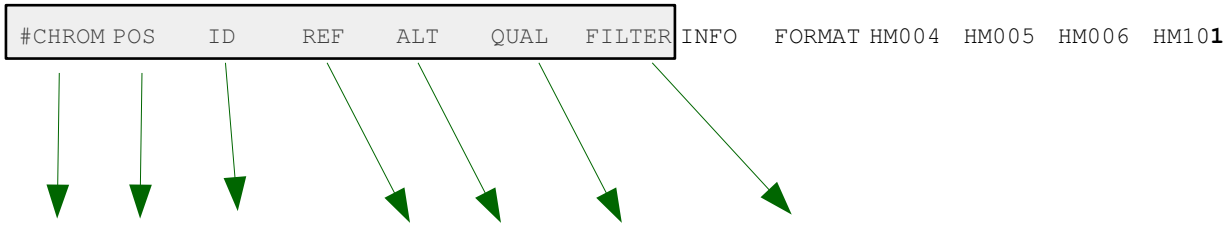
```
#CHROM POS    ID     REF    ALT    QUAL   FILTER INFO   FORMAT HM004  HM005  HM006  HM101
```

```
chr7   157    chr7_157     T      G      8510.72      PASS
AC=178;AF=0.78;AN=214;BaseQRankSum=0.998;DP=800;Dels=0;FS=25.28;HaplotypeScore=2.187;InbreedingCoeff=0.2401;MLEAC=199;MLEAF=0.754;MQ=37.
62;MQ0=0;MQRankSum=-4.652;QD=13.68;ReadPosRankSum=0.716    GT:AD:DP:GQ:PL      ./.:.:.:.:.  1/1:0,6:6:12:143,12,0
1/1:0,3:3:9:102,9,0 0/0:54,1:54:99:0,132,1529
```

```
chr7   261    chr7_261     T      C      4731.37      PASS
AC=176;AF=0.71;AN=212;BaseQRankSum=4.85;DP=749;Dels=0;FS=78.113;HaplotypeScore=1.6073;InbreedingCoeff=0.4641;MLEAC=176;MLEAF=0.638;MQ=37
.66;MQ0=2;MQRankSum=-5.928;QD=20.48;ReadPosRankSum=4.061   GT:AD:DP:GQ:PL      1/1:0,2:2:6:54,6,0  1/1:0,11:11:24:267,24,0
1/1:2,4:6:3:29,3,0  0/0:70,2:70:99:0,195,2207
```

```
chr7   553    chr7_553     G      T      10708.8      PASS
AC=277;AF=0.813;AN=310;BaseQRankSum=4.498;DP=1298;Dels=0.01;FS=342.229;HaplotypeScore=2.0697;InbreedingCoeff=0.3771;MLEAC=257;MLEAF=0.70
6;MQ=36.34;MQ0=0;MQRankSum=-12.248;QD=16.65;ReadPosRankSum=6.033  GT:AD:DP:GQ:PL      1/1:0,4:4:9:82,9,0  1/1:0,5:5:12:135,12,0
1/1:0,7:7:15:162,15,0      0/0:118,8:120:99:0,253,3653
```

Anatomy of a VCF data line




```
AC=176;AF=0.71;AN=212;BaseQRankSum=4.85;DP=749;Dels=0;FS=78.113;HaplotypeScore=1.6073;InbreedingCoeff=0.4641;MLEAC=176;MLEAF=0.638;MQ=37.66
;MQ0=2;MQRankSum=-5.928;QD=20.48;ReadPosRankSum=4.061      GT:AD:DP:GQ:PL      1/1:0,2:2:6:54,6,0  1/1:0,11:11:24:267,24,0
1/1:2,4:6:3:29,3,0  0/0:70,2:70:99:0,195,2207
```

Anatomy of a VCF data line

```
#CHROM POS     ID      REF     ALT     QUAL    FILTER INFO    FORMAT HM004  HM005  HM006  HM101
```

```
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each ALT allele, in the same order as listed">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##INFO=<ID=BaseQRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt Vs. Ref base qualities">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth; some reads may have been filtered">
##INFO=<ID=DS,Number=0,Type=Flag,Description="Were any of the samples downsampled?">
##INFO=<ID=Dels,Number=1,Type=Float,Description="Fraction of Reads Containing Spanning Deletions">
##INFO=<ID=FS,Number=1,Type=Float,Description="Phred-scaled p-value using Fisher's exact test to detect strand bias">
##INFO=<ID=HaplotypeScore,Number=1,Type=Float,Description="Consistency of the site with at most two segregating haplotypes">
##INFO=<ID=InbreedingCoeff,Number=1,Type=Float,Description="Inbreeding coefficient as estimated from the genotype likelihoods per-sample when compared against the Hardy-Weinberg expectation">
##INFO=<ID=MLEAC,Number=A,Type=Integer,Description="Maximum likelihood expectation (MLE) for the allele counts (not necessarily the same as the AC), for each ALT allele, in the same order as listed">
##INFO=<ID=MLEAF,Number=A,Type=Float,Description="Maximum likelihood expectation (MLE) for the allele frequency (not necessarily the same as the AF), for each ALT allele, in the same order as listed">
##INFO=<ID=MQ,Number=1,Type=Float,Description="RMS Mapping Quality">
##INFO=<ID=MQ0,Number=1,Type=Integer,Description="Total Mapping Quality Zero Reads">
##INFO=<ID=MQRankSum,Number=1,Type=Float,Description="Z-score From Wilcoxon rank sum test of Alt vs. Ref read mapping qualities">
##INFO=<ID=QD,Number=1,Type=Float,Description="Variant Confidence/Quality by Depth">
##INFO=<ID=RPA,Number=.,Type=Integer,Description="Number of times tandem repeat unit is repeated, for each allele (including reference)">
##INFO=<ID=RU,Number=1,Type=String,Description="Tandem repeat unit (bases)">
##INFO=<ID=ReadPosRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt vs. Ref read position bias">
##INFO=<ID=STR,Number=0,Type=Flag,Description="Variant is a short tandem repeat">
```

```
chr7   261    chr7_261     T      C      4731.37      PASS
```

```
AC=176;AF=0.71;AN=212;BaseQRankSum=4.85;DP=749;Dels=0;FS=78.113;HaplotypeScore=1.6073;InbreedingCoeff=0.4641;MLEAC=176;MLEAF=0.638;MQ=37.66
;MQ0=2;MQRankSum=-5.928;QD=20.48;ReadPosRankSum=4.061      GT:AD:DP:GQ:PL      1/1:0,2:2:6:54,6,0  1/1:0,11:11:24:267,24,0
1/1:2,4:6:3:29,3,0  0/0:70,2:70:99:0,195,2207
```

Anatomy of a VCF data line

```
#CHROM POS     ID      REF     ALT     QUAL    FILTER INFO    FORMAT HM004  HM005  HM006  HM101

##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=AD,Number=.,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="Normalized, Phred-scaled likelihoods for genotypes as defined in the VCF specification">


chr7    261     chr7_261        T       C       4731.37         PASS

AC=176;AF=0.71;AN=212;BaseQRankSum=4.85;DP=749;Dels=0;FS=78.113;HaplotypeScore=1.6073;InbreedingCoeff=0.4641;MLEAC=176;MLEAF=0.638;MQ=37.66

;MQ0=2;MQRankSum=-5.928;QD=20.48;ReadPosRankSum=4.061       GT:AD:DP:GQ:PL      1/1:0,2:2:6:54,6,0  1/1:0,11:11:24:267,24,0

1/1:2,4:6:3:29,3,0  0/0:70,2:70:99:0,195,2207
```

PL values are phred-scaled likelihood scores, normalized such that the most likely genotype will have a score of 0. The approximate likelihoods are **10^(-PL/10).**

For PL values of 54,6,0 (genotypes of HM004) the likelihoods would be

10^(-54/10) approx. 4E-06

10^(-6/10) approx. 0,25

10^(0) approx. 1

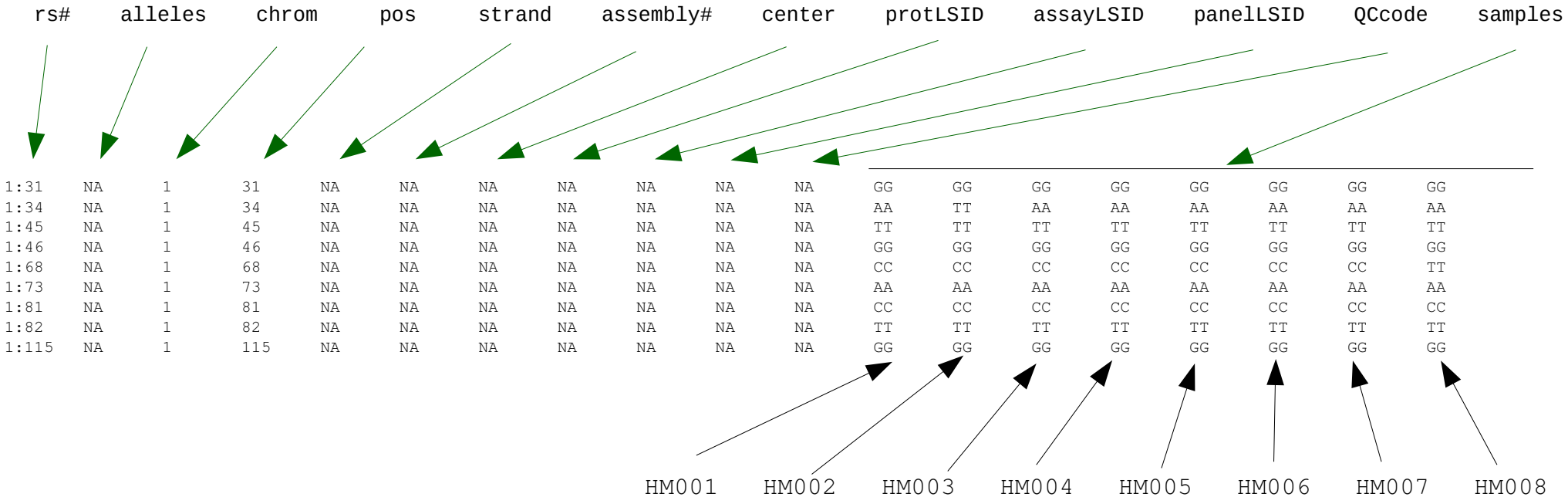
The HapMAP Format

- Tab-delimited file
- The Hapmap file format consists of 11 columns plus one column for each sample genotyped..
- The first row contains the header labels, and each additional row contains all the information associated with a single SNP

- NO documentation inside
- Does NOT give informations of data/tools/parameters used

- Variant lines represent a position on the genome
  - Example : 850,000 SNPs onto 300 accessions :
    - a file with 850,001 lines and 311 columns

The HapMAP Format



Also possible to use the standard IUPAC code (single bit) as following:

Genotype	AA	CC	GG	TT	AG	CT	CG	AT	GT	AC
Code	A	C	G	T	R	Y	S	W	K	M

The PLINK formats – PLINK flat files (MAP/PED)

The fields in a MAP file are:

- Chromosome number[integer]
- SNP ID[string]
- SNP genetic position (cM)[float] may be unknown
- SNP physical position (bp)[integer]

Each SNP must have a unique physical position.
All the SNPs must be ordered by physical position

MAP file - Example

Example of a MAP file of the standard PLINK format:			
21	rs11511647	0	26765
X	rs3883674	0	32380
X	rs12218882	0	48172
9	rs10904045	0	48426
9	rs10751931	0	49949
8	rs11252127	0	52087
10	rs12775203	0	52277
8	rs12255619	0	52481

PED file can be SPACE or TAB delimited.
Each line corresponds to a single individual.
The first 6 columns are:

- Family ID [string]
- Individual ID [string]
- Father ID [string]
- Mother ID [string]
- Sex [integer]
- Phenotype [float]

Columns 7 & 8 code for the observed alleles at SNP1,
Columns 9 & 10 code for the observed alleles at SNP2,
and so on.
Missing data are coded as "0 0"

This file should have N lines and 2L+6 columns,
where N and L are the numbers of individuals and SNPs contained in the dataset respectively.

PED file - Example

FAM1	NA06985	0	0	1	1	A	T	T	T	G	G	C	C	A	T	T	T	G	G	C	C
FAM1	NA06991	0	0	1	1	C	T	T	T	G	G	C	C	C	T	T	T	G	G	C	C
0	NA06993	0	0	1	1	C	T	T	T	G	G	C	T	C	T	T	T	G	G	C	T
0	NA06994	0	0	1	1	C	T	T	T	G	G	C	C	C	T	T	T	G	G	C	C
0	NA07000	0	0	2	1	C	T	T	T	G	G	C	T	C	T	T	T	G	G	C	T
0	NA07019	0	0	1	1	C	T	T	T	G	G	C	C	C	T	T	T	G	G	C	C
0	NA07022	0	0	2	1	C	T	T	T	G	G	0	0	C	T	T	T	G	G	0	0

The PLINK formats – PLINK Binary files (BED/BIM/FAM)

- The binary PLINK format contains the same information as the flat file PLINK format but in a compressed and significantly more efficient form.

BED File

- The BED files are encoded in binary format.
- Binary PED (BED) files.

BIM File

Example of a BIM file of the binary PLINK format:

21	rs115116470	26765	A	T
X	rs3883674 0	32380	C	G
X	rs122188820	48172	T	T
9	rs109040450	48426	A	T
9	rs107519310	49949	C	T
8	rs112521270	52087	A	C
10	rs127752030	52277	A	A
8	rs122556190	52481	G	T

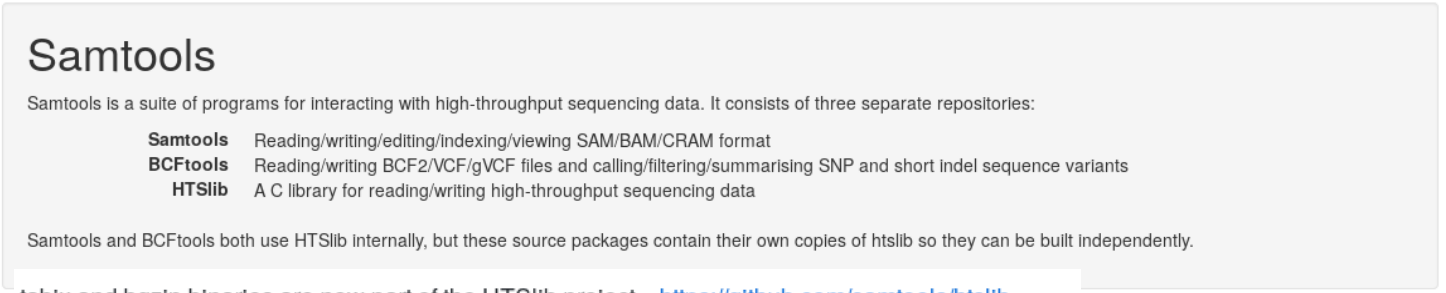
FAM files

- **The fields in a FAM file are**
- Family ID
- Sample ID
- Paternal ID
- Maternal ID
- Sex (1=male; 2=female; other=unknown)
- Affection (0=unknown; 1=unaffected; 2=affected)

- Note: The **FAM file** is just the first six columns of the **PED file**

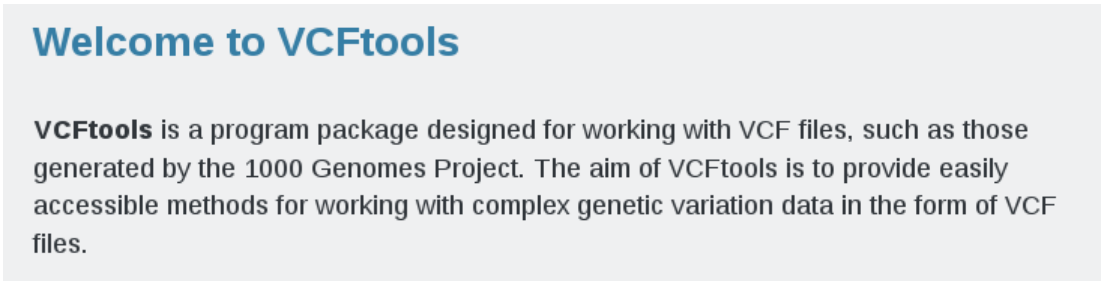
Working with formats, converting between formats

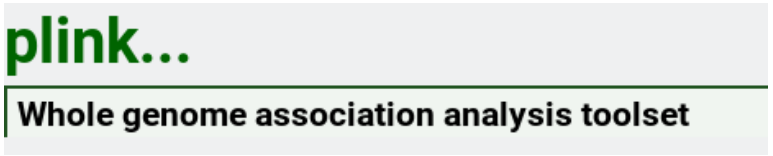
- FASTA ↔ FASTQ
- SAM / BAM
- VCF / BCF
- PED,MAP
- BED, BIM, FAM
- ...

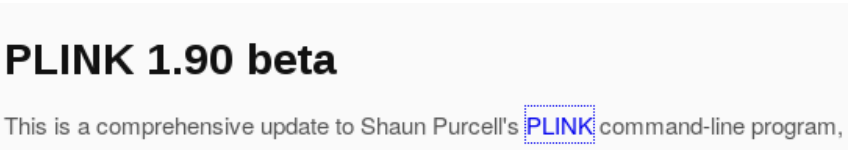


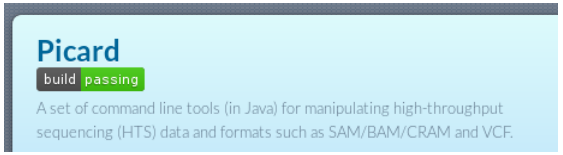
tabix and bgzip binaries are now part of the HTSlib project. https://github.com/samtools/htslib

BamTools is a project that provides both a C++ API and a command-line toolkit for reading, writing, and manipulating BAM (genome alignment) files.







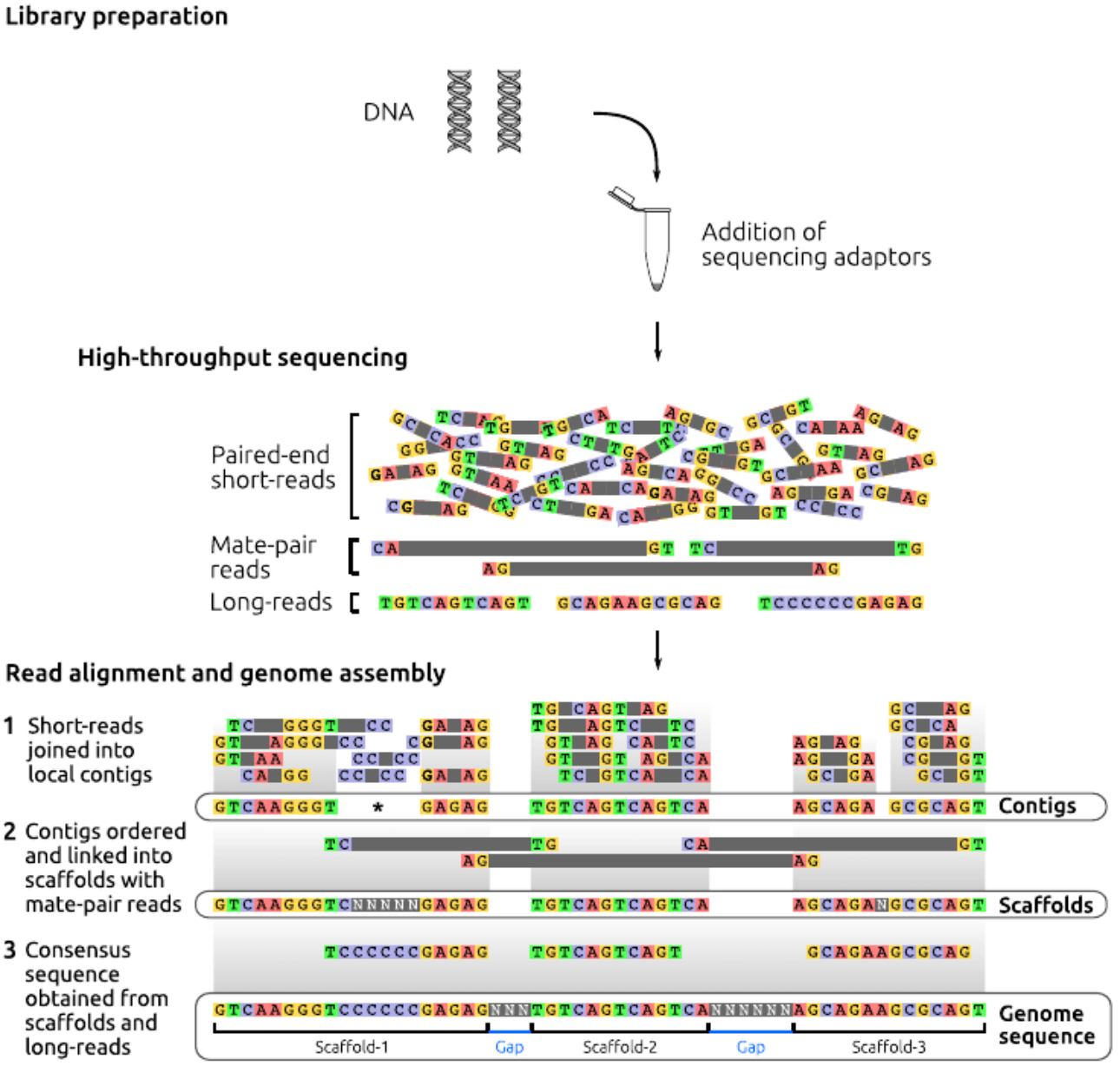




short introduction in bio-informatics

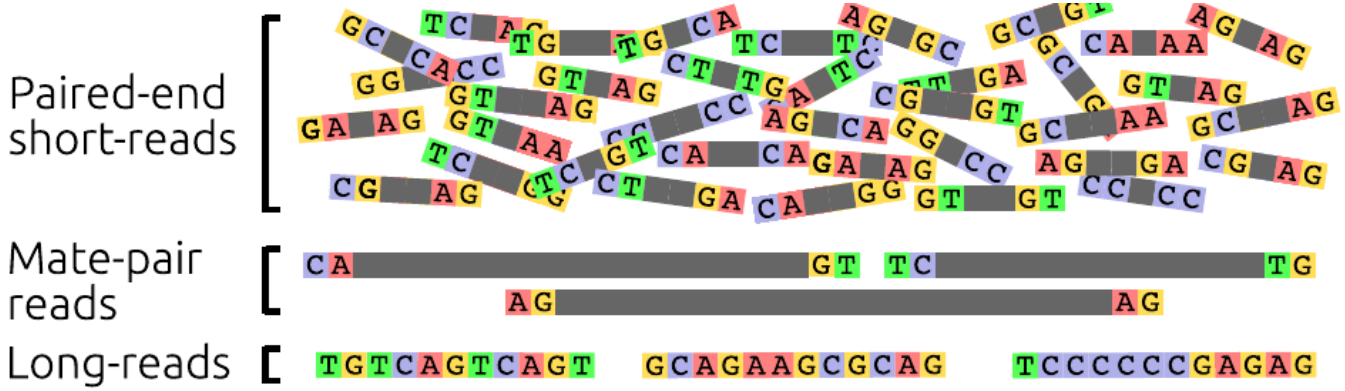
- Detecting SNPs using 'whole genome resequencing'
  - Current tools
  - File/data formats
- Using 'reduced-representation' sequencing libraries (GBS, RAD-Seq , ...) of multiplexed samples
  - GBS with a reference genome
  - GBS w/o a reference genome : UNEAK and STACKS pipelines
- File/data formats for handling SNPs data for subsequent analyses
- quick view : principle of *de novo* sequencing
- What can I do with this genetic data ?

De novo whole-genome assembly



Fuentes-Pardo et al. (2017)
Mol Ecol 26: 5369–5406.

De novo whole-genome assembly

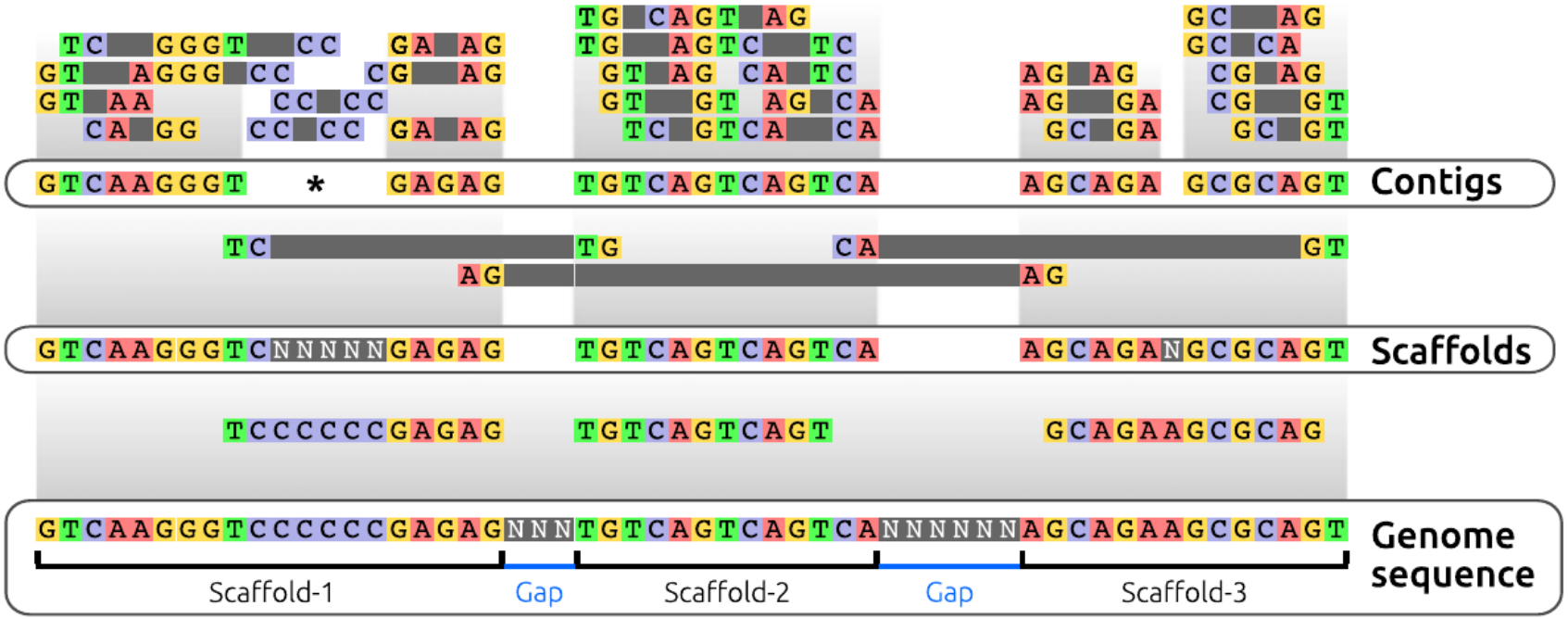


Read alignment and genome assembly

1 Short-reads joined into local contigs

2 Contigs ordered and linked into scaffolds with mate-pair reads

3 Consensus sequence obtained from scaffolds and long-reads



Computing resources in Toulouse



GENOTOUL BIOINFORMATICS HOME

The GenoToul bioinformatics facility is part of the Genotoul GIS. It has been set up in 2000. Since 2009, it is one of the 13 IBISA bioinformatics platforms. Since 2008, the plateform collaborates with the local genomic platform and processes huge volumes of data produced by second and third generation of sequencers and makes them available to biologists (ng6).

EQUIPMENT

- A computer farm : about 5000 cores (INTEL-2014, AMD-2012), 34 Tera Byte memory (3TB on a SMP machine), Infiniband interconnection (QDR), parallel file system (GPFS)
- Web servers and virtual machines hosting infrastructure
- More than 1Peta Byte disk space

SERVICES

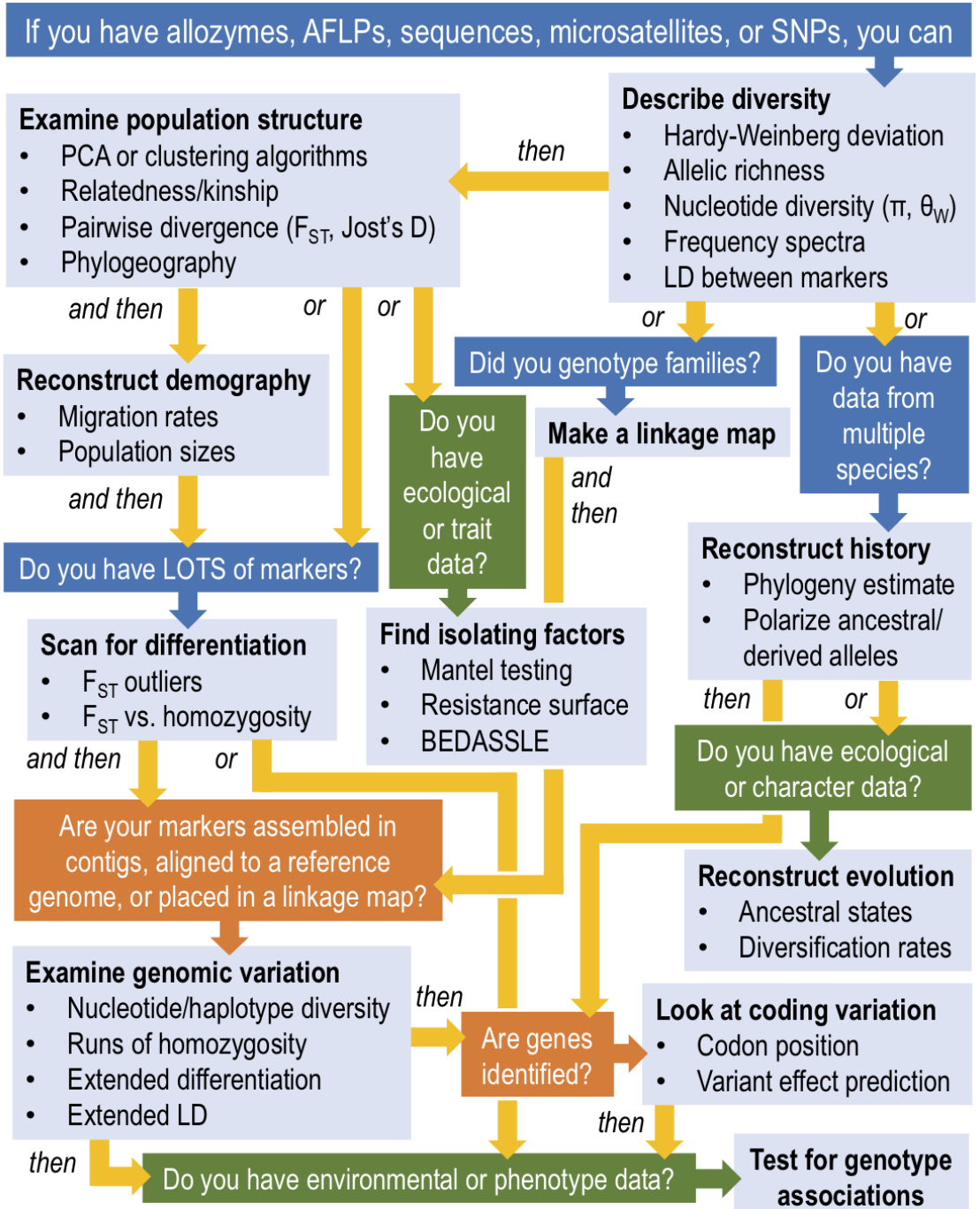
- Access to public biological banks
- Access to generic and specific bioinformatics software pieces
- Access to web resources
- Projects (Web/VM) hosting (ask for a project hosting)
- Training

Use this link to create your user account. All questions about technical issues can be sent using one of the Ask for forms. The platform staff can help you to process your data or to develop specific databases or software packages. For any specific request

short introduction in bio-informatics

- Detecting SNPs using 'whole genome resequencing'
  - Current tools
  - File/data formats
- Using 'reduced-representation' sequencing libraries (GBS, RAD-Seq , ...) of multiplexed samples
  - GBS with a reference genome
  - GBS w/o a reference genome : UNEAK and STACKS pipelines
- File/data formats for handling SNPs data for subsequent analyses
- quick view : principle of *de novo* sequencing
- What can I do with this genetic data ?

What can I do with this genetic data?



INP ENSAT
Institut National Polytechnique, Ecole Nationale Supérieure Agronomique de Toulouse (INP-ENSAT)
cnrs

The Molecular Ecologist
www.molecularecologist.com
wp.me/p4OZed-2Rb

ВИР